Expected Returns and Large Language Models *

Yifei Chen[†]
Booth School of Business
University of Chicago

Bryan Kelly[‡]
Yale University, AQR Capital
Management, and NBER

Dacheng Xiu[§]
Booth School of Business
University of Chicago and NBER

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[†]Address: 5807 S Woodlawn Avenue, Chicago, IL 60637 USA. Email: yifei.chen@chicagobooth.edu.

[‡]Address: 165 Whitney Ave, New Haven, CT 06511 Email: bryan.kelly@yale.edu.

[§]Address: 5807 S Woodlawn Avenue, Chicago, IL 60637, USA. Email: dacheng.xiu@chicagobooth.edu.

Abstract

We use large language models (LLMs), including ChatGPT and the LLaMA family, to extract embeddings from financial news and evaluate their ability to predict cross-sectional stock returns. LLM-based news signals contain incremental information beyond traditional technical predictors, such as return reversal, and firm-level characteristics, and they substantially outperform simpler text representations, including bag-of-words and Word2Vec. Portfolios formed on LLM-implied expected returns deliver economically meaningful Sharpe ratios after transaction costs and are not driven by look-ahead bias. The return dynamics exhibit short-horizon news momentum consistent with underreaction: predictability persists for several days among small stocks but dissipates quickly for large stocks, in line with limits to attention and arbitrage. Decomposing news into economically interpretable topics reveals pronounced heterogeneity in predictive strength, with analyst-related news generating the most profitable signals. In contrast, word-based models perform better when applied to alerts and headlines to full articles, suggesting that LLMs extract deeper semantic information from complete texts. AI-generated summaries balance contextual richness and structure and yield the strongest predictive performance. Using data from 16 global equity markets and news written in 13 languages, we show that these findings are robust across markets and linguistic environments.

Keywords: Artificial Intelligence, Natural Language Processing, Large Language Models, Return Prediction, BERT, LLaMA, ChatGPT

1 Introduction

Economic text is continuously produced by human writers striving to understand and forecast economic phenomena. In recent decades, the finance literature has begun to extract information from certain text sources like the financial news press, regulatory filings, and social media. However, the research agenda of improving economic models through text mining is at an early stage. Research has thus far examined only a limited portion of market-relevant textual data, often focusing on a single specialized data source at a time (e.g., the front page of *The Wall Street Journal*, or the "risk factor" section of 10-K filings). Furthermore, within each data source, text information is often represented in rudimentary ways, such as a dictionary-based sentiment score and "bag of words (BoW)."

There are compelling reasons for the historically limited use of text data in financial applications. The lack of regular structure makes it far more difficult to work with than standard numerical data sets. Language is an extremely nuanced information encoding scheme. As a result, complex models are necessary to faithfully decode information contained in text. Technological barriers to entry exclude researchers who lack the specialized skill sets necessary to operate such models, while high computational costs may exclude other researchers who possess the requisite skills but face research funding constraints.

Therefore, recent textual analysis in finance and economics is the tip of the iceberg. Text remains an underexploited data source for understanding asset markets. The challenges of textual analysis today portend an exciting research agenda tomorrow, in which economists study ever larger text corpora and refine their ability to elicit information from that text.

In this paper, we model stock-level expected returns using information from news article text following a two-step procedure. In Step 1, we derive detailed, high-dimensional, numerical representations of text using large language models (LLMs). Step 2 uses these numerical representations in econometric return prediction models.

1.1 Step 1: LLM Embeddings to Represent Text

LLMs have significantly advanced the state-of-the-art in text modeling. They are massive neural networks pre-trained on extensive and diverse text datasets, encompassing a wide range of sources and topics. Their architectures emphasize awareness of terms in the context of other terms, including consideration of long-range dependencies among terms.

The key advantage of using LLM embedding in Step 1 of our pipeline lies in the model pre-training: a specialized research team performs the Herculean feat of estimating a general-purpose language model with an immense number of parameters, trained on vast corpora including full-text books, large swathes of internet text, and other large-scale textual repos-

itories. But for each LLM, this feat is performed *just once*. Once trained, the estimated LLM is made available for broader distribution, enabling non-specialist researchers to leverage its capabilities for various downstream applications without requiring expertise in model training or large-scale computation. Conceptually, an LLM functions as a powerful prior over language and world knowledge: its large-scale pre-training, conducted by a small group of specialists, encodes rich contextual structure that downstream applications can directly leverage as a plug-and-play solution.

Due to their architecture and large-scale pretraining, LLMs extract substantially richer semantic and contextual information from text than traditional BoW methods.

Beneath these LLM accomplishments lies a neural network architecture. An LLM maps each token into a high-dimensional numerical vector at its last hidden layer. To obtain a representation for an entire text, the model aggregates these token-level representations into a single fixed-length vector, commonly referred to as a document embedding. In practice, this aggregation is implemented either by taking the hidden state of a designated summary token (e.g., a special classification token) or by pooling across token embeddings (such as mean pooling). The resulting embedding encodes the semantic and contextual information extracted from the full text and, in a chatbot setting, conditions subsequent text generation.

Researchers can obtain token-level embeddings directly through open-source implementations or APIs. The result of our Step 1 text representation is a numerical data matrix X of dimension $D \times P$, where D is the number of documents and each row contains the P-dimensional numerical vector embedding of those documents.

1.2 Step 2: Econometric Return Prediction With Embeddings

In Step 2, the embedding matrix X serves as an input to an econometric model designed to forecast stock returns. This acts as a computationally inexpensive "fine tuning" step—akin to fitting a linear "head" on frozen LLM embeddings—so the task-specific mapping is learned without retraining the LLM. The embeddings are treated as fixed features, and all learning occurs in the predictive layer.

Our baseline specification is linear with shrinkage to mitigate overfitting in the high-dimensional feature space. In addition, we consider nonlinear alternatives that allow for richer interactions among embedding dimensions, including tree-based methods and neural networks trained to predict returns. These models assess whether nonlinear transformations of the embeddings provide incremental predictive power relative to the linear benchmark.

1.3 Empirical Findings

Our central empirical finding is that LLM embeddings extract economically relevant information from financial news that is incorporated into stock prices gradually rather than instantaneously. Although financial news is publicly available, much of it is redundant, qualitative, or costly to interpret. As a result, prices need not immediately reflect the full implications of news, even absent differential access. We show that LLM-based representations isolate the informative components of news that predict short-horizon returns and shed light on the dynamics of information assimilation.

Using the two-step framework outlined above, we first encode news articles using pre-trained LLM embeddings and then map these embeddings into expected returns using econometric prediction models.

We examine six large-scale pre-trained LLMs: BERT (developed by Google), RoBERTa (by Meta), and the LLaMA model family, including LLaMA, LLaMA2, and LLaMA3 (by Meta). In addition, we analyze text-embedding-3-large (by OpenAI), an embedding-focused counterpart to ChatGPT, and refer to this model as "ChatGPT" throughout.[1] As benchmarks, we also consider three models based on simpler text representations: the Loughran–McDonald Master Dictionary (LMMD)[32], Word2Vec (by Google), and SESTM [26].

Relative to existing approaches in the finance literature, LLM-based models deliver substantial improvements in out-of-sample return predictability. Importantly, the predictive content of LLM-based signals is not subsumed by traditional technical predictors, such as short-term reversal, nor by firm-level characteristics.

These predictive signals translate into economically large and robust portfolio performance. A daily long–short strategy that buys stocks in the highest predicted-return quintile and sells those in the lowest quintile achieves annualized Sharpe ratios of 3.0, 3.7, 4.0 (4.4, 4.6), and 4.8 for BERT, RoBERTa, LLaMA3 (LLaMA2, LLaMA), and ChatGPT, respectively. These results compare favorably with Sharpe ratios of 2.6 for SESTM, 2.7 for Word2Vec, and 2.3 for LMMD. While our baseline analysis relies on linear predictive models, we show that more flexible downstream algorithms can further enhance performance. Importantly, these gains remain robust after accounting for realistic transaction costs.

We also demonstrate that these findings generalize internationally. Applying a unified prediction framework to 15 non-U.S. equity markets and news written in 12 languages, we find that LLM-based models consistently outperform word-based alternatives across coun-

[1]The key advantage of text-embedding-3-large is that its embeddings are accessible through OpenAI's API, which is not the case for ChatGPT. Access to embeddings is essential for our modeling design and for the replicability of our analysis.

tries and linguistic environments. The predictive advantage of LLM embeddings persists in non-English settings, indicating that they capture transferable economic information rather than language-specific artifacts.

We address concerns about look-ahead bias and information leakage through multiple complementary exercises. These include extending the sample beyond the training cutoff dates of major language models, masking firm identifiers and time-sensitive content, and conducting strictly out-of-sample evaluations. Across these tests, the predictive performance of LLM-based signals remains largely unchanged, supporting the interpretation that the documented predictability reflects genuine information in news text rather than inadvertent access to future outcomes.

The economic nature of this predictability is best characterized as news-driven underreaction. We document short-horizon return continuation following news arrival: returns drift in the direction implied by news for several days before dissipating. This pattern is distinct from the well-known short-term reversal effect in daily stock returns and indicates delayed, but directionally correct, price adjustment rather than overreaction. The effect is strongly heterogeneous across firms. News-induced momentum is concentrated among small stocks and dissipates quickly for large stocks, consistent with limits to attention and arbitrage rather than behavioral overshooting.

We further show that the speed of information incorporation depends on both the timing and the structure of news. Focusing on Refinitiv "news alerts," which correspond to breaking news and consist solely of short headlines, we find that return predictability is strong but short-lived. In this setting, LLM-based signals continue to generate economically meaningful returns, but their advantage over simpler word-based methods narrows. This evidence suggests that when information is extremely timely and textual content is sparse, speed of dissemination dominates differences in language model capacity. In contrast, for full news articles, which contain richer narrative context, LLMs substantially outperform word-based approaches and generate more persistent return predictability. Consistent with this interpretation, we find that AI-generated summaries of news articles often provide especially effective inputs for return prediction. Summaries balance informational richness with structure, delivering stronger and more stable performance than either headlines or full article text alone.

To further interpret the economic content underlying the model's signals, we decompose traded news into semantically coherent topics using LLM embeddings and hierarchical clustering. This topic-based partition reveals substantial heterogeneity across information channels. Categories that contain sharp forward-looking expectation updates—such as sell-side recommendation changes, earnings disclosures, and explicit valuation revisions—generate the

strongest and most persistent long–short returns. In contrast, more narrative or qualitative themes—such as corporate strategy discussions or general management commentary—tend to exhibit weaker and less persistent performance.

1.4 Related Literature

Our paper is the first to use textual embeddings from LLMs to model expected asset returns. In the current financial text mining landscape, most methods use supervised machine learning models customized to specific tasks such as predicting returns [26, 23], predicting volatility [36], or predicting macroeconomic activity [27, 7]. Unlike our embeddings-based approach, the most common choice for text representation in these supervised models is BoW, which collapses a document into a high-dimensional vector that counts occurrences of unique terms. In some cases, the numerical representation stops here [e.g. 23, 27]. In other cases, the numerical representation is refined further, for instance by estimating bespoke sentiment libraries through correlation screening [26] or through a dimensionality-reducing topic model [7].

BoW representations and their derivatives are limited by the fact that text information is conveyed solely through the frequency of term occurrences. It sacrifices nearly all information that is conveyed by word ordering and contextual relationships between terms. Second, the high dimensionality of BoW representations[2] introduces statistical inefficiencies because a downstream econometric models must estimate a large number of parameters to utilize these terms, even though most terms carry little to no information. More subtly, models based on BoW are *corpus specific*. For example, [7] estimate their topic model using articles from *The Wall Street Journal*, despite the fact that many of their topics reflect general language structures that can be better inferred if additional text outside of their data set is incorporated.

Subsequent to our work, embeddings have been used for a number of economic applications outside of return prediction:[3] from asset representations derived from data on investment positions of various market participants [15], to construct global networks from firms' business descriptions [5], to gauge historical country-level financial sentiment from a large corpus of books [25], to model product demand from multimodal data [2], and to relate news to asset valuations [41], among others.

An emergent literature uses LLMs via their text generation or "prompt-response" functionality, rather than through their underlying neural network embeddings. The leading

[2]The number of unique words in a corpus often number in the tens of thousands or more, and if the BoW includes "multigrams" (phrases of multiple words) the number of terms can be orders of magnitude larger.

[3]A draft of our paper was first circulated in 2022.

example is [31], which prompts ChatGPT with news headlines and a question of whether the headline is good, bad, or neutral for a stock. They then document significant profitability of a daily trading strategy that buys stocks with good news and sells stocks with bad news.[4] Prompt-based and embedding-based modeling approaches are highly complementary. On the one hand, prompt-based analysis occurs in the space of language thus enjoys a high degree of interpretability. On the other hand, prompt-based methods are limited by the researcher's ability to engineer meaningful prompts, and LLM responses can be highly sensitive to small changes in the prompt and typically incorporate randomness even in controlled settings (which can hinder replicability). Perhaps most importantly, the scope of information included in the LLMs response is dictated by the prompt, which is typically narrow by design to target specific text content sought by the prompt engineer.

Embeddings-based models take a different tack. Rather than capturing a specific information type dictated by the prompt, the embedding seeks to capture general and comprehensive information content in the text (Step 1 in the two-step procedure outlined earlier). In addition, embeddings are deterministic and avoid subjective biases of the prompt engineer. Embeddings can be flexibly adapted to downstream tasks such as return prediction which serves as a computationally inexpensive "fine tuning" step (Step 2 in the two-step procedure). More concretely, while one may prompt an LLM with a specific question like "is this good news about stock i?", our embedding approach will instead represent the entire article text as a numerical vector, and the subsequent regression step will seek the best use of that comprehensive embedding for the task of return prediction. If there are predictable economic phenomena outside of the prompt's scope, the embedding approach still has some ability to capture it. We emphasize again however that the approaches are complementary and can be viewed as different points along the bias-variance tradeoff. Prompt-based prediction models are more specific and structured (thus more biased and lower variance), while embeddings-based predictions are more comprehensive and less structured (less biased and higher variance).

The rest of the paper is organized as follows. Section 2 introduces the LLMs and other approaches that we compare. Section 3 presents an empirical analysis of stock-level news and return prediction in the US and international markets using these methods. Section 4 shows economic interpretation of text-based trading strategies. Section 5 concludes. The Internet Appendix contains supplementary tables and figures. These provide additional details on international market summary statistics, the timing of trading implementation, the news-data masking procedure, portfolio performance across model scale variants, the effect of text truncation on predictive performance, and representative text excerpts for each topic

[4]Other prompt-based uses of LLMs in economics include [6], [12], [33], and [**?**], among others.

category.

2 The Textual Analysis Framework

This section describes our approach to textual analysis. We begin with an overview of text embeddings derived from LLMs. Then we describe how we use these embeddings to model expected stock returns.

LLMs are a class of deep learning models for text data. These models are defined not only by their large number of parameters—often in the billions or more—but also by their transformer-based architectures that are trained on large and diverse text corpora. Designed to learn language patterns and representations in a general-purpose manner, LLMs are the greatest examples to date of "artificial intelligence" and exhibit broad adaptability across a wide range of tasks.

2.1 Extracting Embeddings from LLMs

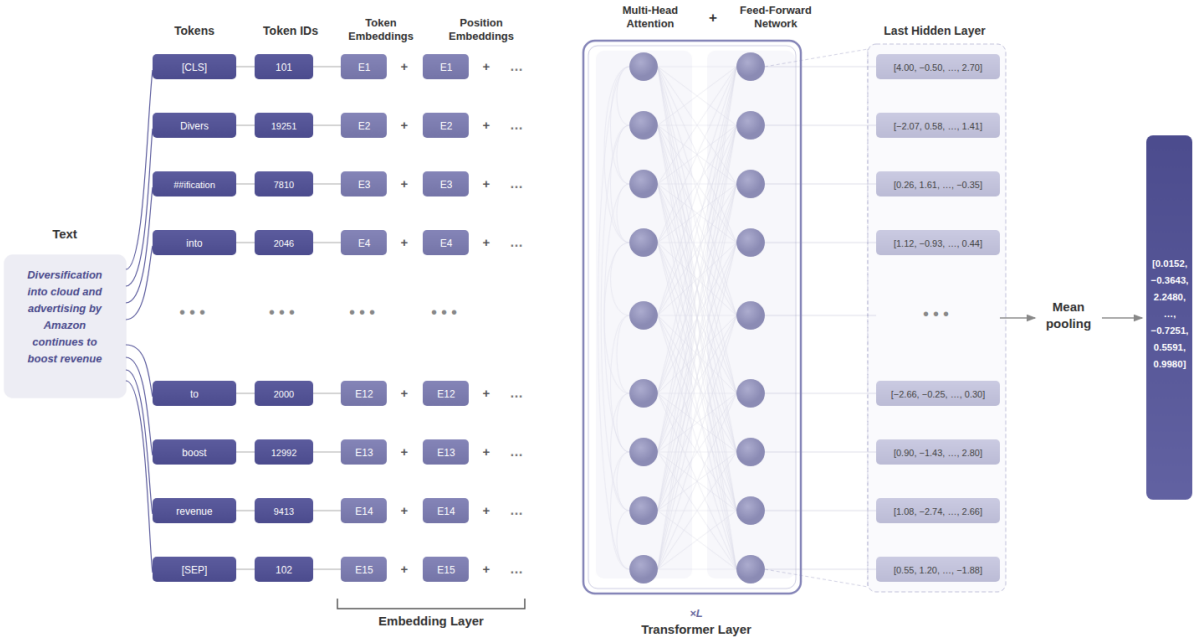
For every article we analyze, we construct a single numerical vector representation which we refer to as an article embedding. In this section we outline the procedure for computing article embeddings from a given LLM.

The basic unit of analysis in an LLM is a "token." The first model component in an LLM is a "tokenizer," which divides article text into these basic units. Tokens can take the form of complete words, word segments, and characters (e.g., punctuation marks), depending on the rules of the tokenizer.

The article, now represented as a sequence of tokens, is then input into the neural architecture of the LLM. All of the models we study are variations of attention-based transformer models [3, 34, 45]. The critical aspect of the transformer is that it evaluates tokens jointly and has the flexibility to capture dependencies among tokens even when they occur far from each other in the sequence. The output of the transformer architecture is a fixed-length embedding vector *for each token*. The token-specific embeddings are computed jointly across the entire sequence, so that each token's representation reflects information from all other tokens in the article. Leveraging pre-trained attention weights that encode statistical dependencies in language, the transformer generates context-dependent embeddings that capture each token's meaning conditional on the full text.

Each LLM has an embedding of fixed length P (which can differ across LLMs). An article with A tokens processed through the LLM produces an $A \times P$ matrix of embeddings. Our final article embedding is simply the $P \times 1$ equally weighted average of embeddings for all

Figure 1: Illustration of LLM's Embedding Pipeline



Note: This figure illustrates the tokenization and embedding pipeline in a transformer-based language model. The input text is first tokenized and mapped into vector embeddings, which are then processed through stacked transformer layers. The resulting contextualized token representations are aggregated (e.g., via average pooling in our implementation) to obtain a fixed-length document embedding. This document-level representation serves as the input to downstream prediction tasks.

tokens in the article.[5] This general pipeline is illustrated in Figure 1.

While our use of LLMs share this general pipeline, they differ in the details of their tokenizers, in their transformer architectures, and in their "context window" (i.e., the maximum length of the input token sequence). Newer models have more efficient tokenization methods, often encoding the same phrase with fewer tokens, which allows an LLM to process longer documents or capture more context within the same computational budget. In terms of architecture, older LLMs are encoder-only models that capture bidirectional context and optimized for understanding natural language. Newer architectures tend to be decoder-only models, which are autoregressive in nature and well-suited for text generation and open-ended tasks such as dialogue. As Figure 1 illustrates, distinctions between tokenizers, architectures, and other model details do not affect our generic empirical pipeline, but are background model features that ultimately determine differences across LLMs in their return predictability.

[5]Mean pooling over token embeddings is also used in [29, 40]; [43] provide comparative evidence against max pooling and last-token summaries.

2.2 From Embeddings to Expected Returns

We use econometric prediction models to translate news article embeddings into expected stock returns over short horizons. Our main model specification is a linear panel regression that expresses returns as a function of article-level features. Let $x_{i,t}$ denote the $P \times 1$ embedding vector extracted from a news article associated with stock i at time t. The model is defined as:

$$\mathbb{E}(r_{i,t+1} \mid x_{i,t}) = x'_{i,t}\theta, \tag{1}$$

where $r_{i,t+1}$ represents the next period return.

Inspired by the empirical framework in [18], we train this pooled panel regression model across all stocks and time periods in the training set and evaluate its predictive performance on out-of-sample data. To control overfitting in the high-dimensional embedding space, we use ridge regularization. The linear formulation provides a transparent and tractable benchmark for assessing the predictive power of various text embeddings.

While the linear specification provides a useful starting point, it may lack sufficient flexibility to accurately map text embeddings to future returns. To capture nonlinear dependencies between text and returns (including interactions among different entries of the embedding vector), we also consider nonlinear models of the form

$$\mathbb{E}(r_{i,t+1} \mid x_{i,t}) = f(x_{i,t}), \tag{2}$$

where $f(\cdot)$ denotes a nonlinear prediction function that we estimate with various machine learning techniques including random forest and neural networks.

We focus at first on one-day-ahead return predictability. Our objective in this analysis is to assess whether information in financial news systematically anticipates subsequent price movements, thus providing insight into the short-horizon efficiency of financial markets.[6]

We train all models (both LLM-based and the benchmark models below) using an annually updated rolling window. Each window spans an eight-year period, with the first six years allocated for training and the following two for validation. The subsequent one-year

[6] An important extension is to study longer-horizon return predictability. One natural approach is to redefine the prediction target as multi-day returns and to aggregate embeddings over rolling windows of past news, allowing models to capture lower-frequency information such as changes in firm outlook or macroeconomic exposure. The embedding-based approach is particularly well suited for this setting, as temporal aggregation is simple and scalable. By contrast, prompting-only workflows must concatenate multiple days of raw articles for each forecast, which quickly runs into context-window constraints when articles about many firms are included or when the horizon spans weeks to months. We leave a systematic investigation of these extensions to future research.

dataset is reserved for out-of-sample testing. Consequently, the out-of-sample period extends from 2004 to 2019, covering 16 years in total.

2.3 From Expected Returns to Portfolios

To evaluate the out-of-sample predictive performance of LLMs in economic terms, we design trading strategies that use models' estimated expected returns to make investment decisions. Our trading strategies follow a standard procedure of constructing a zero-net-investment portfolio that takes long positions in the top quintile of stocks with the highest expected returns on each date and short positions in stock in the bottom quintile expected returns. We consider both equal-weighted and value-weighted portfolios schemes. Equal weighting assigns identical weights to all stocks, providing a broad test of predictive power across firms of different sizes. Value weighting assigns weights proportional to market capitalization, placing greater emphasis on larger and more liquid firms and reflecting economically scalable investment opportunities.

Additionally, in line with a similar choice made by [26], we forecast open-to-open returns, which closely mirrors how quantitative strategies are implemented in practice. Large portfolio managers typically do not adjust positions instantaneously as firm-specific news arrives, owing to technological and liquidity constraints. The overnight interval thus serves as a natural window to aggregate information accumulated during the prior day. Overnight news is processed and incorporated at the next market open, whereas news released during the previous trading session may already be reflected in prices—making our specification conservative.

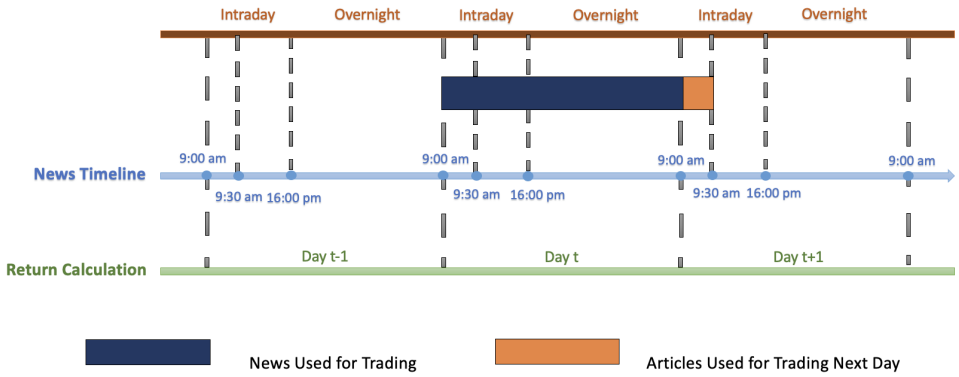
Following the finance text-analysis literature, we exclude articles published between 9:00 a.m. and 9:30 a.m. Eastern Time for U.S. markets and shift them to the next day's input set. This convention reflects realistic operational frictions: traders need time to ingest data, update or refit models, and compute positions before the opening auction. The adjustment facilitates execution when liquidity is deepest at the open while preserving the full information set, avoiding the need to discard potentially informative content. We illustrate the timing convention in Figure 2.

2.4 The LLMs We Study

Over the course of our research the landscape of large language models has evolved rapidly.[8] Computing embeddings from LLMs is cost intensive, especially with newer and larger models,

[8] An earlier draft circulated in 2022, at which time our analysis focused on Open Pre-trained Transformer (OPT) released by Meta.

Figure 2: Trading Timeline



Note: This figure illustrates the news and trading timeline. To ensure a conservative implementation, news released between 9:00 and 9:30 AM EST is shifted to the next trading day. News arriving on day t determines portfolio positions formed at the market open on day $t+1$, which are held throughout the trading day and rebalanced at the subsequent market open.

Table 1: Summary of Language Models Used in This Study

Model	**Dir.**	**Vocab Size**	**Params**	**Emb. Dim.**	**Multi- ling.**	**Release Date**	**Cut-off Date**
BERT Large	Bi	30K	345M	1,024	No	Oct 2018	Oct 2018
RoBERTa Large	Bi	50K	355M	1,024	No	Jul 2019	Jul 2019
LLaMA (13B)	Uni	32K	13B	5,120	Yes	Feb 2023	Feb 2023
LLaMA2 (13B)	Uni	32K	13B	5,120	Yes	Jul 2023	Jul 2023
LLaMA3 (8B)	Uni	128K	8B	4,096	No	Apr 2024	Mar 2023
ChatGPT	Uni	–	–	3,072	Yes	Nov 2022	Jan 2022

Note: This table summarizes the key characteristics of the large language models evaluated in our study. **Model** refers to the specific version used in our experiments. **Dir.** (Direction) indicates whether the model processes text using bidirectional attention (Bi), which considers both preceding and following context, or unidirectional attention (Uni), which considers only preceding context. **Vocab Size** denotes the number of distinct text tokens the model recognizes. **Params** shows the number of model parameters, providing a rough measure of model scale. **Emb. Dim.** (Embedding Dimension) refers to the dimensionality of each token's vector representation produced by the model. **Multi- ling.** indicates whether the model supports multiple languages. **Release Date** is the public release date of the model, and **Cut-off Date** refers to the most recent point in time covered by the data used for model training.[7]

thus it is not feasible to examine every leading LLM that is available. We focus on a set of six models that are reasonably representative of the state-of-the-art. These span both a variety of neural network architectures and differ in terms of training data, training method, model size, languages, and accessibility, as summarized in Table 1. Our selection is designed to capture major developments in the evolution of LLMs while maintaining computational tractability for large-scale empirical implementation. At the same time, the chosen models remain economically relevant for downstream prediction tasks and fall within our available

computing budget.

To guide the discussion, we describe each model in terms of six key characteristics: (1) architecture and training objective, (2) number of parameters, (3) support of multiple languages, (4) context window, (5) embedding dimension, and (6) accessibility (open source or API).

2.4.1 BERT

We begin with BERT [Bidirectional Encoder Representations from Transformers 10], which serves as our baseline, and introduce other models in sequence, highlighting how each one differs from or builds upon its predecessors. BERT represents a foundational advance in the development of language models. Released in 2018, it introduces a new approach to understanding tokens in the context of surrounding language. The core goal of BERT is text understanding rather than text generation. Because of this, it models word predictions bi-directionally—i.e., in the context of language that comes both before and after the token being forecasted. This bi-directional contextual approach results in large improvements across a range of language tasks relative to BERT's predecessors.

BERT leverages the transformer architecture introduced by [45], a model design that processes entire sequences of text in parallel and uses an attention mechanism [3, 34] to identify relationships between words regardless of their position. This design, which has since become standard in modern LLMs, replaced earlier approaches that processed text sequentially and had difficulty capturing long-range dependencies.

BERT is trained as a masked language model, meaning that some words in a sentence are hidden, and the model learns to predict them based on the surrounding (bi-directional) context. It is also trained using next sentence prediction, whose goal is to determine if two candidate sentences are consecutive in their original text. The version we study for English language text is BERT-large. It is trained on BookCorpus (roughly 11,000 English language books) and English Wikipedia, and has 340 million parameters. It uses a fixed vocabulary of around 30,000 tokens and can take in sequences of up to 512 tokens, and produces embeddings with dimension 1,024. For non-English text, distinct BERT models have been pre-trained for each language we consider. We use the BERT-base model corresponding to the specific language for languages other than English. Specific model names for each language are list in Table IA2 of the internet appendix. BERT models are open source.

2.4.2 RoBERTa

RoBERTa [Robustly Optimized BERT Pretraining Approach, 30] builds directly on BERT by modifying the training process. It removes the next sentence prediction objective, uses a more dynamic approach to masking words, and expands the BERT training data by a factor of five through its inclusion of CommonCrawl internet text. These changes consistently improve performance across a range of benchmark language tasks.

RoBERTa retains BERT's architecture but uses a different tokenizer. Its vocabulary is larger than BERT at around 50,000 distinct tokens. The base model includes 125 million parameters and, like BERT, is trained on English-language data and publicly available. For non-English text we use the multilingual variant, XLM-RoBERTa, which shares the same architecture but is trained on text from 100 languages. Its vocabulary is substantial (around 250,000 tokens) to accommodate the broader range of linguistic structures, and the base multilingual model has 270 million parameters. Like BERT, RoBERTa is open source and has a context window of 512 tokens and embedding dimension 1,024.

2.4.3 LLaMA Family

The family of LLaMA models [Large Language Model Meta AI 44], comprises improvements in scale, data quality, and efficiency relative to the models above. LLaMA epitomizes the rapid evolution of LLMs in recent years, which we leverage by studying three versions of LLaMA to help understand how the continual improvement of this model impacts its usefulness for return prediction. Different LLaMA vintages have multiple versions with different numbers of parameters, see the Internet Appendix A. All LLaMA models are open source and trained on multilingual data, and remain among the most capable openly available language models to date.

All three versions are transformer-based and are trained to predict the next word in a sequence using only preceding words.[9] This uni-directional attention architecture is particularly effective for tasks that involve generating or completing text and sequential reasoning, and is a major divergence from the bi-directional framework of BERT. Each version of LLaMA introduces specific enhancements:

- LLaMA (February 2023) is trained on 1.4 trillion tokens, uses a vocabulary of around 32,000 tokens. We use the 13 billion parameter version. It's context window is 2,048 tokens and its embedding dimension is 5,120.

[9]To improve reliability and safety, models undergo subsequent supervised fine-tuning in which they are trained on human-annotated examples to perform specific tasks.

- LLaMA2 (July 2023) expands to 2 trillion training tokens that have been filtered for quality. We again use the 13 billion parameter version. The context window is extended to 4,096 tokens and its embedding dimension is also 5,120.

- LLaMA3 (April 2024), increases the vocabulary to 128,000 tokens and uses improved tokenization. In this case we use the 8 billion parameter version. It supports a context window of 8,000 tokens and its embedding dimension is 4,096.

2.4.4 ChatGPT

We use ChatGPT in the form of OpenAI's text-embedding-3-large model. It uses the GPT architecture, a family of transformer models trained for uni-directional prediction. The GPT family has evolved significantly over time. The original model, introduced by [39] uses 117 million parameters, while later releases like GPT-4 remains proprietary and is believed to use over a trillion parameters.

In our study, we do not access the full GPT model directly. Through the text-embedding-3-large API (released in January 2024), we access GPT embeddings that are optimized for tasks such as semantic similarity and information retrieval. The exact model size and tokenizer details are undisclosed. It has a context window of 8,191 tokens and an embedding dimension of 3,072.

2.5 Non-LLM Benchmarks

To benchmark the performance of embeddings-based return predictions, we study three models from the literature that use non-LLM supervised machine learning to forecast returns.

2.5.1 Loughran-McDonald

The first benchmark is a dictionary-based methodology from [32] (LMMD henceforth). These authors construct the Loughran-McDonald Master Dictionary using human domain knowledge regarding financial text, manually assigning sentiment scores to individual tokens. To apply the LMMD method, we first condense each article into its BoW representation using a standard tokenizer. Then, following the procedure in [32], we construct an article-level scaled sentiment score by aggregating the sentiment scores for all tokens in the article. In this case, the article sentiment score serves as the (scalar) article embedding $x_{i,t}$.

2.5.2 SESTM

Our second benchmark is the SESTM model for predicting returns with text data [26]. SESTM begins with a BoW tokenization of articles. It then trains a machine learning architecture to simultaneously estimate a low-dimensional dictionary of tokens that most strongly predict returns, and the sign prediction coefficients associated with those tokens.[10]

Finally, for each article, it aggregates tokens using SESTM's estimated coefficients to produce a scalar article-level sentiment score (which, like LMMD, is treated as a scalar article embedding $x_{i,t}$). Unlike pre-trained models, SESTM is estimated on our Refinitiv data sample in a recursive manner that allows us to make out-of-sample forecasts without look-ahead bias. The Internet Appendix B discusses the structure and training of SESTM in more detail.

SESTM is a fully specified model that avoids the need for a pre-defined dictionary or domain-specific knowledge. Its simplicity, theoretical foundations, and powerful empirical performance make SESTM a valuable BoW-based benchmark for our LLM analysis.

2.5.3 Word2Vec

In the intervening era between BoW-based models and before LLMs, textual analysis centered on vector representations of words, and is epitomized by Word2Vec [37]. Rooted in distributional semantics [19, 13], the rationale for word vectors are based on the concept that "you shall know a word by the company it keeps." Using large training sets, word vectors are trained to summarize the meaning of words in light of frequent neighboring words. However, unlike the token embeddings in modern LLMs, Word2Vec produces a single static vector that does not change based on the surrounding words. In other words, Word2Vec produces unconditional word embeddings, rather than context-conditional word embeddings like those in LLMs.

We use pre-trained word vectors for multiple languages from the fastText platform.[11] Word vectors are 300-dimensional, and we average the vectors for all words in an article to obtain an article-level Word2Vec embedding.[12]

To close our discussion of benchmarks, we note that BoW methods (like Loughran-

[10] [26] use the sign of the three-day return around firm news as a classification target. For comparability across methods, we instead use next-day returns as the prediction target.

[11] FastText, developed by [4], was chosen due to its multilingual support. Although we adopted the fastText package, we consistently use the term "Word2Vec" for convenient reference.

[12] For English, we specifically used the *wiki-news-300d-1M* model [38], comprising one million word vectors trained on Wikipedia 2017, the UMBC web-base corpus, and the statmt.org news dataset (totaling 16 billion tokens). For non-English languages, word vectors were similarly derived from Common Crawl and Wikipedia data [17].

Table 2: Summary Statistics of US News Articles and Alerts

Raw Articles			Articles Tagged with Single Stock			Articles With	Excluding Very	Excluding
RTRS	3PTY	Total	RTRS	3PTY	Total	Returns	Short & Long	Duplicates
6,366,019	4,843,867	11,209,886	2,863,166	4,123,823	6,986,989	4,755,247	4,123,279	3,038,025

Raw Alerts	Alerts Tagged with Single Stock	Alerts With	Excluding
RTRS	RTRS	Returns	Duplicates
4,976,374	4,054,683	3,286,003	2,935,852

Note: This table reports the sample size after applying successive filters related to tagging a single stock, matching with return data, ensuring appropriate length, and removing redundancy, for news articles (top panel) and news alerts (bottom panel).

McDonald and SESTM) as well as shallow distributional embeddings like Word2Vec, require meticulous data pre-processing. We follow the procedure outlined in [26] and use the natural language toolkit (NLTK) in Python for pre-processing across all languages. This extensive procedure includes converting text to lowercase, expanding contractions (e.g., "haven't" to "have n't"), performing lemmatization (reducing words to their root forms), tokenization, and removal of pronouns, proper nouns, punctuation, special symbols, numbers, and common stop words like "and," "the," and "is." By contrast, modern LLMs ingest raw text as sequences of tokens and do not require lemmatization or stop-word removal, largely eliminating the burden of extensive data cleaning.

3 Empirical Analysis

3.1 Data and Pre-processing

We source our news text data from Refinitiv. This dataset encompasses global news from their RTRS (Reuters News) and 3PTY (Third Party News) datasets from January 1996 to June 2019. For US firms, the new documents fall into two categories: *articles* and *alerts*. Articles include both a headline and a body of text, providing a comprehensive narrative of various firm events. Alerts are real-time updates of what Refinitiv flags as breaking news, and alerts contain only a headline. Third party and international news databases consist solely of articles. We integrate this news text data with US equity data (returns and market capitalizations) from CRSP and international equity data from EIKON (Datastream).[13]

We implement several filters for our analysis. First, we retain only those articles and

[13] After the first draft of the paper was posted, we obtained an extension of the same dataset for U.S. news coverage through May 2022. The extended dataset contains only U.S. news articles and does not include U.S. alerts or international news coverage. This additional sample is used exclusively for robustness checks, including tests related to potential forward-looking bias.

alerts "tagged" as associated with a single stock (according to Refinitiv's article metadata) and for which the tagged stock has non-missing open-to-open returns around the article date. We remove articles with fewer than 100 characters or exceeding 100,000 characters (this filter is not applied to the set of shorter alert documents).[14] Next we impose a filter to exclude duplicate (and near-duplicate) documents. Specifically, using the BoW representations of news documents, we exclude those with a cosine similarity of 0.8 or higher compared to another article from the prior five business days.[15]

Table 2 summarizes post-filtering document counts for US stocks. The volume of news data is substantial, containing over approximately six million documents split evenly between articles and alerts. The most impactful filter is requiring documents to be tagged to a single stock. Of the articles that are dropped, the large majority are comprised of general news not tagged to any stock and a minority are tagged to multiple stocks.[16]

Figure 3 shows the volume of filtered articles (left column) and alerts (right column) over time. The top row shows document frequency by year and illustrates a generally increasing pattern in number of documents over time, with document counts rising above the long-term trend around the 2008 global financial crisis. The middle row shows the monthly seasonal patterns in document counts, where we observe cyclical peaks in February, May, August, and November that align with corporate earnings announcement season for most firms (this pattern is especially pronounced for alerts). The lower row shows intra-day seasonal patterns, with most news activity around market open and close, mirroring intraday trading volume seasonalities.

In addition to the US market, we analyze news-based return prediction in international markets including China (Hong Kong), the UK, Australia, Canada, Japan, Germany, Italy, France, Sweden, Denmark, Spain, Finland, Portugal, Greece, and the Netherlands. While data for most countries begin in 1996, aligning with the start of the US sample, some countries have later inception dates due to gaps in news availability.

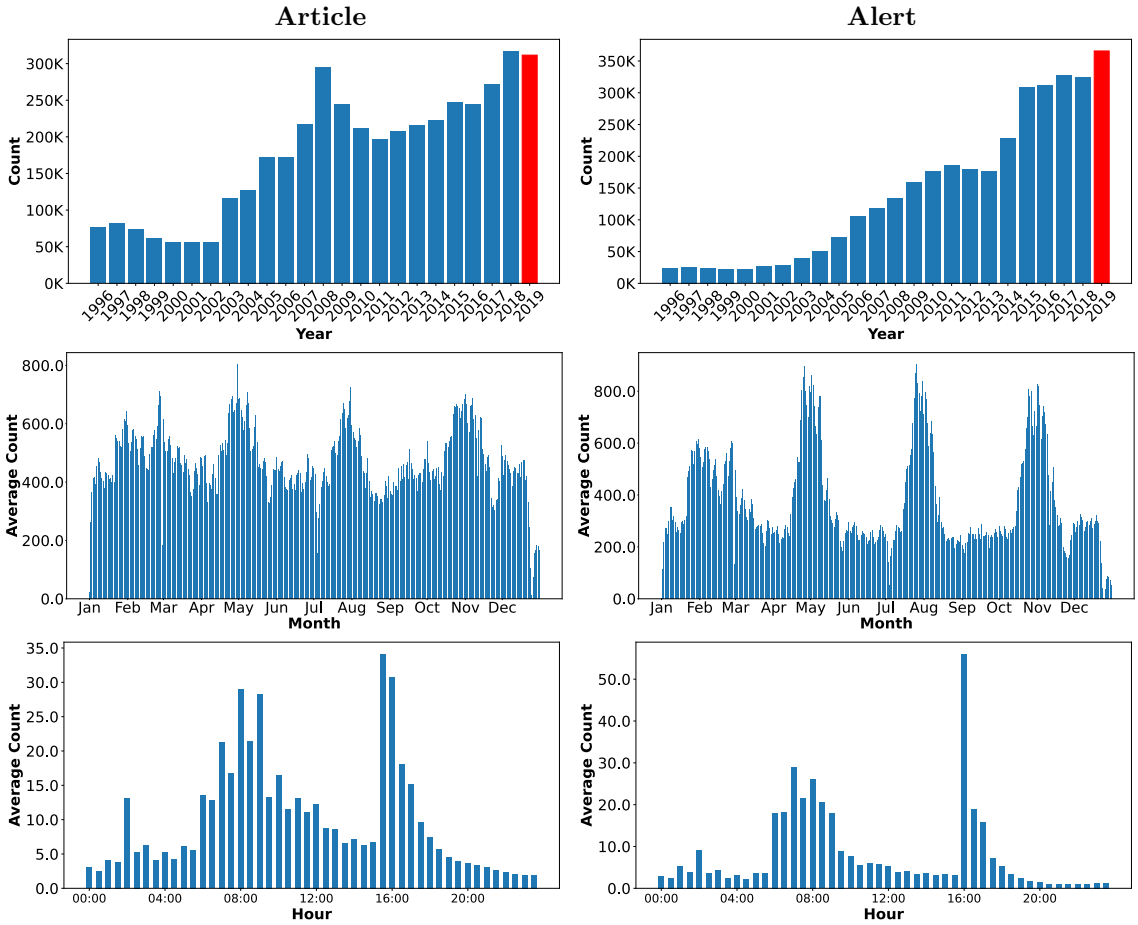
There is substantial cross-country heterogeneity in news coverage. Figure 4 presents the annual news document count for each international market, highlighting pronounced dispersion in article volume, ranging from 3,751 articles in the Netherlands to 571,285 in the UK. In parallel, the average number of stocks with at least one news article in a given month

[14] Excluding very short documents removes noisy items with insufficient content. An upper length cutoff has negligible impact; it simply filters atypically long reports that are not the news we study.

[15] This filter primarily removes articles that are essentially redundant—such as wire updates or minor revisions of previously published stories—so that the retained news reflects genuinely new information rather than repeated coverage of the same events.

[16] We exclude articles tagged to multiple stocks because the information in such an article may be positive for one stock and negative for another (e.g., if the two tagged stocks are competitors). Extending our framework to more carefully accommodate such scenarios is an interesting avenue for further research.

Figure 3: US News Document Counts



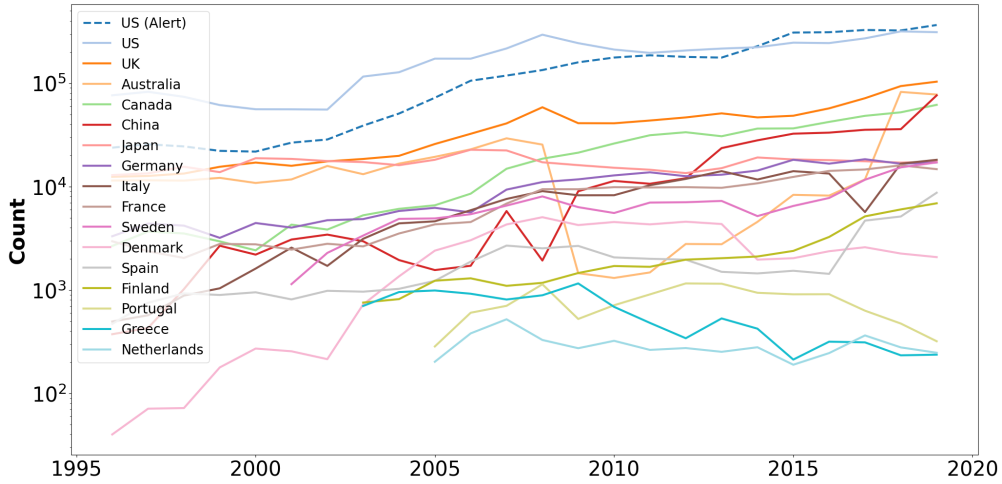
Note: The top panels show the total number of news articles (left) and alerts (right) published annually. Since the sample ends in June 2019, the counts for 2019 are estimated and highlighted in red. The middle panels present the average number of articles and alerts published each month. The bottom panels display the average number of articles and alerts published at different hours of the day.

varies considerably across countries, from 645 in Japan to only 11 in the Netherlands. Detailed summary statistics for the international news and stock market data sets are reported in Tables IA6 and IA7 of the internet appendix.

3.2 Prediction Performance

In Table 3 we evaluate the out-of-sample return prediction performance of each model using cross-sectional correlations between model-based return predictions formed on day t and realized returns on day $t+1$. We report the time series average of daily cross-sectional correlations as well as the pooled stock-day correlations. We further present average cross-sectional correlations for the subset of stocks with market capitalization above the NYSE

Figure 4: News Document Counts By Country



Note: The figure shows annual counts of cleaned and preprocessed news articles across all international markets, with the blue dashed line indicating the number of news alerts for the US equity market.

median, and separately for each calendar year.

All models except the pure word-based approaches (SESTM and LMMD) exhibit consistently positive predictive correlations over the sample. LLM-based models uniformly dominate traditional word-based methods, with ChatGPT and the LLaMA family standing out by achieving average out-of-sample correlations exceeding 2%, thereby outperforming all other specifications.

In Table 4 we conduct pairwise comparisons of methods to assess differences in out-of-sample predictive accuracy using the [11] test. A positive test statistic indicates that the column model outperforms the row model. Bold values denote significance at the 5% level for one-sided test. Because Table 4 involves multiple comparisons, we also evaluate statistical significance using a conservative Bonferroni correction as in [18]. This adjustment divides the significance level by the number of comparisons—setting a one-sided critical value of 2.45 for the 5% level across seven model comparisons. Tests that exceed this stricter threshold are marked with an asterisk (*).

The results reveal a clear hierarchy across model classes. All transformer-based language models decisively outperform the pure word-frequency benchmark LMMD, and the LLaMA family together with ChatGPT also achieve strong and often statistically significant gains over SESTM, as well as over Word2Vec and BERT in most pairwise comparisons. These findings highlight the importance of model scale and architectural advances for return prediction. Within the top-performing group, however, performance differences are modest. Pairwise Diebold–Mariano statistics among LLaMA, LLaMA2, LLaMA3, and ChatGPT are

small and statistically insignificant, suggesting diminishing marginal gains at the frontier and broadly comparable predictive accuracy across the most advanced LLMs.

Table 3: Out-of-sample Return Prediction Performance

	ChatGPT	LLaMA3	LLaMA2	LLaMA	RoBERTa	BERT	Word2Vec	SESTM	LMMD
Mean	2.68	2.45	2.46	2.36	1.88	1.67	1.53	1.39	0.98
Pooled	2.29	2.43	2.36	2.36	1.99	1.66	1.48	1.06	0.37
Over NYSE Median	1.11	1.19	1.24	0.94	0.92	0.60	0.62	-0.06	-0.43
2004	1.42	1.40	0.88	1.92	1.73	1.49	0.87	1.09	0.88
2005	2.07	1.15	1.89	1.72	1.51	1.00	0.87	0.84	1.09
2006	0.92	0.31	0.50	0.61	0.12	0.28	0.32	-0.05	0.98
2007	2.69	2.98	3.11	1.31	0.63	0.78	0.46	0.67	1.14
2008	1.85	2.21	2.17	2.03	1.62	1.29	1.06	0.78	0.07
2009	2.72	2.43	2.73	2.39	1.93	1.81	2.06	1.17	0.84
2010	2.30	1.43	1.29	1.16	1.13	1.42	1.37	1.91	1.36
2011	2.60	0.76	0.61	0.56	0.31	0.77	0.30	1.40	0.51
2012	3.57	3.06	3.04	2.82	2.63	1.90	2.00	1.33	0.78
2013	3.45	2.82	3.10	2.84	2.26	1.96	1.59	1.42	1.58
2014	2.49	3.04	2.96	2.94	2.60	2.13	2.45	2.13	1.03
2015	3.34	3.32	2.90	2.90	2.41	2.06	2.30	1.80	1.31
2016	2.40	2.48	2.50	2.54	1.98	2.21	2.04	1.65	0.61
2017	3.43	3.72	3.69	3.80	2.67	2.25	2.33	2.00	1.20
2018	4.01	4.40	4.23	4.59	3.55	2.72	2.23	2.26	1.08
2019	4.50	4.81	5.03	4.97	4.18	3.66	2.87	2.40	1.39

Note: The table reports the time-series mean of daily cross-sectional correlations (scaled by 100) between predicted and realized daily returns across models. We also report pooled correlations across all stock-days, subsample results for large-cap stocks, and year-by-year averages. The results are reported in percentage points.

Table 4: Comparison of Out-of-sample Predictions using Diebold-Mariano Tests

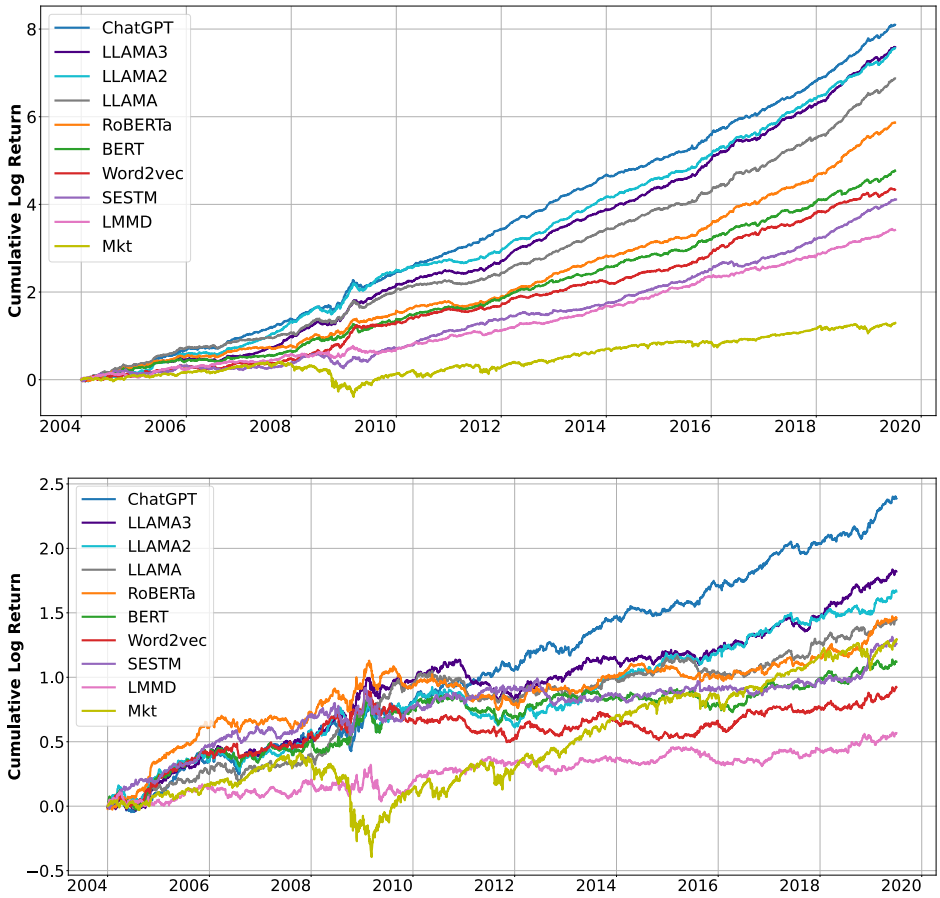
	SESTM	Word2Vec	BERT	RoBERTa	LLaMA	LLaMA2	LLaMA3	ChatGPT
LMMD	**2.109**	**2.095**	**2.097**	**2.098**	**2.106**	**2.106**	**2.107**	**2.106**
SESTM		0.958	1.527	**1.962**	**2.682***	**3.085***	**3.149***	**2.713***
Word2Vec			0.695	0.749	1.516	**1.909**	**1.699**	1.631
BERTLARGE				0.370	1.531	**2.034**	**1.772**	1.642
RoBERTa					**1.847**	**2.030**	**2.138**	1.610
LLaMA						0.350	0.485	0.027
LLaMA2							0.083	-0.280
LLaMA3								-0.307

Note: This table reports pairwise Diebold-Mariano test statistics comparing the out-of-sample prediction performance among the nine models for top and bottom quintile predictions. Positive numbers indicate the column model outperforms the row model. Bold font indicates the difference is significant at 5% level or better for individual tests, and an asterisk indicates significance at the 5% level for 9-way comparisons via our conservative Bonferroni adjustment.

3.3 Portfolio Performance

Following the procedure described in Section 2.3, we evaluate out-of-sample predictive performance in economic terms by implementing a trading strategy that relies exclusively on

Figure 5: Cumulative Returns of Language Model Trading Strategies

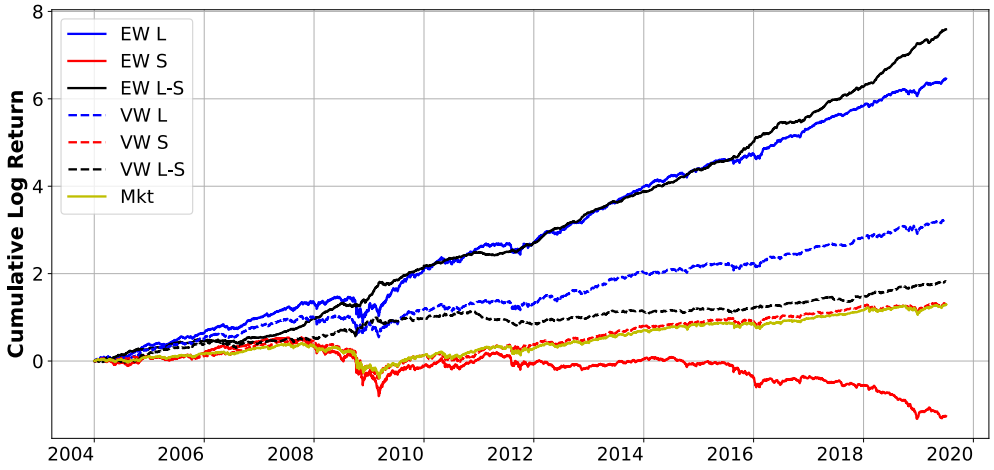


Note: This figure presents the cumulative log returns of US long-short portfolios constructed based on next-period return predictions from ChatGPT, LLaMA3, LLaMA2, LLaMA, RoBERTa, BERT, Word2Vec, SESTM, and LMMD models. The left panel shows equal-weighted strategies and the right panel shows value-weighted strategies. The line labeled "Mkt" represents the cumulative return of the S&P 500.

model-implied expected return forecasts.

Figure 5 plots the cumulative out-of-sample returns of the long–short strategies for all models. The top panel reports equal-weighted results. These strategies avoid major drawdowns, generate positive returns even during the financial crisis, and consistently outperform the market. The smoothness of the return paths reflects strong diversification benefits from trading a large cross section of stocks at equal weights, while the steep upward slope points to small-cap inefficiency and the models' ability to forecast price movements from news text. The bottom panel reports value-weighted strategies, which are substantially more volatile. Because portfolio weights are concentrated in a small number of large-cap stocks, diversification is more limited and returns are driven by fewer positions. Despite this increased volatility, advanced language models such as the LLaMA family and ChatGPT continue to

Figure 6: Detailed Performance Breakdown for LLaMA3 Trading Strategy



Note: This figure compares the out-of-sample cumulative log returns of portfolios sorted on sentiment scores based on the LLaMA3 model. The black, blue, and red colors represent the long-short (L-S), long (L), and short (S) portfolios. The solid and dashed lines represent equal-weighted (EW) and value-weighted (VW) portfolios. The yellow solid line is the S&P 500 return.

outperform the market by a wide margin, indicating that their predictive content remains economically valuable even when portfolio construction emphasizes large-cap stocks. Figure 6 provides a closer look at the LLaMA3 strategy by decomposing cumulative returns into their long and short legs and comparing the equal-weighted and value-weighted implementations.

Table 5 reports a formal analysis of average trading strategy performance for each model. Among all methods, ChatGPT stands out, producing the highest equal-weighted annualized Sharpe ratio of 4.78. The three LLaMA variants follow closely behind; although their performance is slightly lower, they remain in a similar range with Sharpe ratios of 4.59, 4.36, and 4.04. RoBERTa and BERT come next, yielding Sharpe ratios of 3.69 and 3.00. By contrast, the word-based approaches show noticeably weaker results: Word2Vec reaches 2.74, and SESTM attains 2.57.[17]

Value-weighted portfolio performance is uniformly lower, reflecting the heavier weight placed on large-cap stocks, which are generally more informationally efficient. Nonetheless, the relative ranking of model performance remains broadly similar. The newest LLMs (ChatGPT and the LLaMA family) deliver the strongest performance, with Sharpe ratios ranging from 0.89 to 1.34. RoBERTa follows closely at 0.92, with SESTM slightly lower at 0.74. BERT attains a Sharpe ratio of 0.70, and Word2Vec ranks last at 0.60.

Taken together, these results indicate that news articles are substantially stronger pre-

[17]LMMD attains Sharpe ratios of 2.33 (EW) and 0.40 (VW), untabulated to conserve space.

Table 5: Performance of Portfolios Sorted By Return Predictions

	ChatGPT						LLaMA3					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.42	-0.11	0.53	0.24	0.08	0.16	0.43	-0.07	0.50	0.22	0.09	0.12
Std	0.20	0.22	0.11	0.19	0.21	0.12	0.20	0.22	0.11	0.19	0.20	0.11
SR	2.10	-0.48	4.78	1.24	0.37	1.34	2.08	-0.33	4.59	1.11	0.47	1.10
	LLaMA2						LLaMA					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.44	-0.06	0.50	0.23	0.12	0.11	0.39	-0.06	0.45	0.20	0.10	0.10
Std	0.21	0.22	0.11	0.20	0.20	0.11	0.20	0.22	0.11	0.20	0.20	0.11
SR	2.12	-0.26	4.36	1.15	0.60	1.02	1.91	-0.28	4.04	1.02	0.51	0.89
	RoBERTa						BERT					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.36	-0.02	0.38	0.21	0.11	0.10	0.33	0.02	0.31	0.19	0.11	0.08
Std	0.20	0.22	0.10	0.19	0.21	0.11	0.20	0.22	0.10	0.19	0.21	0.11
SR	1.76	-0.11	3.69	1.12	0.53	0.92	1.63	0.08	3.00	1.01	0.54	0.70
	Word2Vec						SESTM					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.32	0.04	0.29	0.21	0.14	0.07	0.28	0.01	0.27	0.22	0.13	0.09
Std	0.20	0.22	0.10	0.19	0.20	0.11	0.20	0.22	0.11	0.19	0.21	0.12
SR	1.60	0.18	2.74	1.08	0.70	0.60	1.36	0.02	2.57	1.12	0.60	0.74

Note: This table presents the performance of equal-weighted and value-weighted quintile long-short portfolios, along with their respective long and short legs, for ChatGPT, LLaMA3, LLaMA2, LLaMA, RoBERTa, BERT, Word2Vec and SESTM models. Sharpe ratios for the short leg are computed on the leg's own P&L; negative values indicate profitable standalone shorts.

dictors of future returns for small stocks. This pattern is consistent with several economic mechanisms. First, small stocks receive less investor attention, so news—whether released overnight or during the previous trading day—may be incorporated less fully or less precisely into prices by the next market open. Second, the underlying fundamentals of small stocks are more uncertain and less transparent, making information more costly to interpret and aggregate into prices. These features may contribute to the higher degree of return predictability observed among smaller firms.

Next, decomposing performance by trade direction shows that both the long and short sides of the strategies are profitable, although the long side typically contributes more to overall returns. For example, with LLaMA3 the equal-weighted long-side Sharpe ratio is 2.08 compared to 0.33 for the short side, and the value-weighted long-side Sharpe ratio is 1.11 compared to -0.47 for the short side. This can be partially attributed to the long side naturally capturing the market equity risk premium, while the short side bears the cost of it. Additionally, it is possible that investors face constraints on short sales, preventing negative news from being fully reflected in prices over short time horizons.

Given the variety of models available, we next examine how different approaches translate textual information into investment strategies. Figure 7 reports a heatmap of pairwise correlations in daily portfolio returns. Correlations among word-based models are relatively low, indicating that these methods extract heterogeneous and weakly related signals from the same news corpus. Correlations between LLM-based portfolios and word-based portfolios are also substantially lower, reflecting fundamental differences in how the two model classes process textual information. In contrast, correlations among LLM-based models are consistently high, suggesting that they converge on similar economic interpretations of news.

Simple examples help illustrate the mechanism. In an article stating that "concerns have been raised" about IAG's post-Brexit ownership structure, word-based models can misclassify the news as positive because "raise" often carries positive sentiment in isolation, while LLMs infer the negative implication from the phrase-level context. A complementary case arises when negative words appear within positive constructions. For instance, an article discussing revenue improvements at INTC notes that "INTC is set to erase all 2018 losses." Word-based models misinterpret this statement as negative due to the token "losses", while LLMs recognize that it describes an improvement in firm performance. Together, these examples highlight that LLMs' ability to process contextual information helps rationalize both the superior performance of LLM-based strategies and their closer alignment in portfolio returns.

We further conduct robustness checks in Table 6 by varying certain implementation details when constructing portfolios. To conserve space, we focus on the performance of the LLaMA3 model. To begin with, we reduce the number of stocks traded each day from the top/bottom quintiles (20%) to deciles (10%). The equal-weighted Sharpe ratio slightly decreases from 4.59 to 4.28, while the value-weighted Sharpe ratio rises from 1.10 to 1.25. We also exclude all news that occurs around earnings announcement days to examine the effect of non-earnings-related news. In doing so, we select again the top/bottom quintile of stocks on each day t, but avoid those whose earning announcements are scheduled on days $t-1$ through $t+1$. The resulting Sharpe ratios—4.30 (EW) and 1.10 (VW), compared to 4.59 (EW) and 1.10 (VW) in the benchmark—indicate that a substantial portion of the predictive content in news is not mechanically driven by earnings announcements. Finally, we evaluate strategy performance when embeddings forecast residual open-to-open returns—computed from CAPM and Fama–French models with factor returns constructed from the intraday Fama–French factors of Aït-Sahalia et al. [1]—instead of total returns. Forecasting residuals yields a modest incremental improvement in portfolio performance.

The Internet Appendix D provides additional empirical analyses that assess the robustness and scope of our main findings. First, we examine the sensitivity of portfolio per-

Figure 7: Correlation of Returns among Different Strategies



Note: The figure presents the correlation of daily portfolio returns among different strategies built on the basis of ChatGPT, LLaMA3, LLaMA2, LLaMA, RoBERTa, BERT, Word2Vec, SESTM, and LMMD models. The correlation of equal-weighed portfolios is shown in the lower triangle, while the correlation of value-weighed portfolios is shown in the upper triangle.

formance to trade execution timing by evaluating open-to-open signals under alternative execution schemes, including VWAP and close-to-close returns, to assess the role of intra-day price adjustment. Second, we study how portfolio performance varies across LLaMA model generations and parameter scales, showing that gains from larger or newer models are modest and non-monotonic. Third, we assess robustness to context length by comparing embeddings constructed from truncated 512-token inputs with those using the full 8K-token context, finding little sensitivity to truncation up to a certain limit. Taken together, these exercises indicate that the qualitative conclusions are broadly consistent with, and reinforce, the results reported in the main text.

3.4 Advanced Machine Learning Models

Having established the robustness of the LLM-based signals and their predictive power under a ridge regression baseline, we next examine whether greater flexibility in the downstream prediction model can further enhance performance when applied to the same LLM-extracted

Table 6: Alternative Models Performance on Daily Portfolio

	BenchMark		Trading Top Decile		w/o. Earning Days	
	Sharpe Ratio	Avg. Ret.	Sharpe Ratio	Avg. Ret.	Sharpe Ratio	Avg. Ret.
EW L-S	4.59	0.50	4.28	0.67	4.30	0.46
EW L	2.08	0.43	2.34	0.50	1.92	0.39
EW S	-0.33	-0.07	-0.72	-0.17	-0.31	-0.07
VW L-S	1.10	0.12	1.25	0.19	1.10	0.12
VW L	1.11	0.22	1.33	0.28	1.05	0.20
VW S	0.47	0.09	0.43	0.09	0.43	0.08
	CAPM Resid.		FF3 Resid.		FF6 Resid.	
	Sharpe Ratio	Avg. Ret.	Sharpe Ratio	Avg. Ret.	Sharpe Ratio	Avg. Ret.
EW L-S	4.58	0.50	4.85	0.52	5.03	0.53
EW L	2.03	0.42	2.10	0.43	2.10	0.43
EW S	-0.37	-0.08	-0.39	-0.08	-0.44	-0.09
VW L-S	0.92	0.10	0.81	0.09	0.73	0.08
VW L	1.03	0.21	0.98	0.20	0.94	0.19
VW S	0.53	0.10	0.56	0.11	0.57	0.11

Note: This table presents the performance of equal-weighted and value-weighted long-short portfolios, including their respective long and short legs, based on the LLaMA3 model. The first panel reports the benchmark LLaMA3 quintile portfolio, while the other two panels show alternative portfolios that either limit daily trading to the top 10% of stocks or exclude firms near their earnings announcement days. The last three columns present the performance of portfolios constructed using CAPM, Fama-French three-factor (FF3), and Fama-French six-factor (FF6) model residuals. All portfolio returns are annualized.

news representations. Using LLaMA3 embeddings throughout, we compare ridge regression to a set of more flexible econometric and machine learning methods—Lasso, random forest, and neural networks—for predicting next-period stock returns.

The ridge and Lasso specifications follow standard penalized linear regressions, with regularization parameters selected via cross-validation.[18] Random forests are implemented as regression forests designed to accommodate the high dimensionality of the feature space, using large ensembles with subsampling and depth constraints chosen via grid search.[19] The neural network specification is a feedforward multilayer perceptron with three hidden layers of decreasing width, trained using stochastic gradient descent with explicit regularization and early stopping.[20]

For each model, we construct quintile portfolios based on predicted cross-sectional returns and compute the corresponding Sharpe ratios, which are reported in Table 7. Lasso

[18]For ridge regression, the penalty parameter is chosen from a log-uniform grid spanning 10^{-1} to 10^{8}, scaled by sample size and feature dimension. For Lasso, the penalty is selected from a log-uniform grid spanning 10^{-4} to 10^{5}.

[19]The random forest consists of 10,000 trees with min samples leaf of 100. Subsample ratio is chosen from $0.3, 0.5, 0.8$. Maximum depth is selected from $10, 20, 30$ and the number of candidate features per split from "$sqrt$", "$P/5$", "$P/10$", "$P/20$".

[20]The network has three hidden layers with 128, 64, and 32 neurons. Each layer uses batch normalization, ReLU activation, and dropout. ℓ_2-regularization is applied to all dense layers, with the penalty selected from $0.1, 0.05, 0.01, 0.005$. Learning rates are chosen from $0.05/\{100, 200, 500, 800, 1000\}$. Training employs early stopping with a patience of five epochs.

Table 7: Advanced Machine Learning Models Performance

	Ridge						LASSO					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.43	-0.07	0.50	0.22	0.09	0.12	0.28	-0.07	0.35	0.16	0.04	0.12
Std	0.20	0.22	0.11	0.19	0.20	0.11	0.18	0.20	0.11	0.18	0.19	0.12
SR	2.08	-0.33	4.59	1.11	0.47	1.10	1.57	-0.34	3.09	0.89	0.22	0.99
	RF						NN					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.52	-0.10	0.62	0.24	0.10	0.14	0.49	-0.11	0.60	0.23	0.05	0.18
Std	0.21	0.22	0.11	0.20	0.21	0.12	0.21	0.22	0.11	0.20	0.20	0.12
SR	2.51	-0.46	5.51	1.21	0.51	1.14	2.32	-0.51	5.26	1.15	0.24	1.58

Note: This table provides a comparative performance of equal-weighted and value-weighted long-short portfolios overall four machine learning models: random forest (RF), Lasso, Ridge, and neural networks (NN).

performs substantially worse than ridge regression, consistent with the findings of [42] that sparsity-inducing methods are dominated by dense models when predictive signals are weak. Motivated by this evidence, we do not consider boosted regression trees and instead rely on ℓ_2 regularization—implemented via explicit penalties or early stopping—when training neural networks, in line with their theoretical analysis and empirical recommendations.

Random forest and neural networks deliver the strongest results overall, with each excelling under different portfolio constructions. Random forest perform best in the equal-weighted specification, achieving a long–short Sharpe ratio of 5.51, while neural networks deliver strong and stable performance across both weighting schemes. In particular, the neural-network value-weighted strategy attains a Sharpe ratio of 1.58, outperforming all competing models in that specification.

Overall, while nonlinear machine learning methods deliver strong portfolio performance, their relative success depends on modeling choices and portfolio construction. To remain conservative and transparent, and to emphasize the contribution of LLM-based representations rather than the choice of prediction algorithm, we therefore carry our main analysis through ridge regression.

3.5 Transaction Costs Analysis

In the preceding sections, our evaluation focused on establishing the economic relevance and magnitude of the predictive content embedded in each model, abstracting from transaction costs. To provide a first assessment of practical implementability, we emphasized value-weighted portfolio strategies, which naturally concentrate on larger and more liquid stocks and thus mitigate implementation frictions associated with text-based return prediction.

An alternative approach to assessing practical viability is to evaluate equal-weighted

Table 8: Transaction Cost: LLaMA3

Panel A: Articles

		Gross		Net	
γ	Turnover	Return	Sharpe Ratio	Return	Sharpe Ratio
0.10	16.71	3.75	1.93	1.49	0.77
0.20	34.44	6.20	2.85	1.56	0.72
0.30	52.47	8.35	3.46	1.28	0.53
0.40	70.61	10.29	3.87	0.78	0.29
0.50	88.79	12.06	4.16	0.12	0.04
0.60	106.99	13.72	4.35	-0.66	-0.21
0.70	125.20	15.30	4.47	-1.53	-0.45
0.80	143.45	16.81	4.55	-2.46	-0.67
0.90	161.84	18.28	4.59	-3.46	-0.87

Panel B: Alerts

		Gross		Net	
γ	Turnover	Return	Sharpe Ratio	Return	Sharpe Ratio
0.10	17.97	5.65	2.13	3.33	1.26
0.20	36.59	9.41	3.07	4.70	1.53
0.30	55.23	12.85	3.70	5.74	1.65
0.40	73.82	16.04	4.13	6.54	1.68
0.50	92.35	19.02	4.42	7.14	1.66
0.60	110.82	21.82	4.60	7.57	1.60
0.70	129.23	24.48	4.72	7.87	1.52
0.80	147.64	27.04	4.79	8.05	1.43
0.90	166.17	29.49	4.81	8.13	1.33

Note: This table reports the transaction cost analysis for the LLaMA3 model. The analysis is divided into two panels: Panel A presents results based on articles, while Panel B reports results based on alerts. Portfolios are constructed as equally weighted long-short EWCT portfolios, with the EWCT parameter denoted by γ. Average returns are reported in basis points per day, and Sharpe ratios are annualized. Portfolio average daily turnover is calculated as $\frac{1}{2T}\sum_{t=1}^{T}\left(\sum_i \left| w_{i,t+1} - \frac{w_{i,t}(1+y_{i,t+1})}{1+\sum_j w_{j,t}y_{j,t+1}} \right|\right)$.

portfolio strategies after explicitly accounting for transaction costs. To measure net performance, we construct a transaction cost model that allows for differential trading costs across stocks of different sizes. Following [14], we assume daily transaction costs of 10 basis points (20 basis points) per 100% turnover for large (small) stocks, where size is defined relative to the NYSE market capitalization 20th percentile.

In addition, we incorporate a turnover-reduction mechanism following [26]. Specifically, we construct trading signals using exponentially weighted calendar time (EWCT), which restricts portfolio rebalancing to a fixed fraction in each period. This procedure amounts to assigning exponentially decaying weights to stocks based on the recency of their appearance in the news, thereby extending effective holding periods and further mitigating transaction costs.

Table 8 evaluates the impact of transaction costs on LLM-based trading strategies using the LLaMA3 model and ridge regressions. The table reports results for equally weighted long–short portfolios, with the EWCT smoothing parameter γ governing portfolio turnover,

such that only a γ fraction of the portfolio is rebalanced each day.

Panel A shows that strategies based on full news articles are highly sensitive to trading frictions. As γ increases, turnover rises sharply, and transaction costs materially reduce net performance. At high turnover—for example, $\gamma = 0.6$—trading costs more than offset the gains from news-based return prediction, leading to negative net returns and Sharpe ratios. Lower values of γ substantially reduce trading activity and preserve economically meaningful net performance, albeit at the cost of lower gross returns.

Panel B reports results for alert-based strategies and reveals substantially greater robustness to transaction costs. Across the full range of γ, alert-based portfolios continue to deliver positive net returns and Sharpe ratios, even at relatively high turnover levels. For instance, at $\gamma = 0.9$, the gross Sharpe ratio reaches 4.81, while the net Sharpe ratio remains 1.33 after transaction costs. More generally, net performance is maximized at an intermediate turnover level around $\gamma = 0.4$, highlighting the importance of turnover control for practical implementation.

We next compare transaction cost performance across language models and predictive specifications. Table 9 summarizes transaction cost–adjusted Sharpe ratios across all language models. Several clear patterns emerge. First, consistent with the LLaMA3 results, alert-based strategies substantially outperform article-based strategies in terms of both net returns and Sharpe ratios across all turnover levels. Moreover, the turnover–net performance trade-off documented earlier persists across models: higher turnover raises gross performance but also amplifies trading costs, leading to a deterioration in net Sharpe ratios.

Second, large language models dominate traditional word-based approaches after transaction costs. In article-based strategies, classical text representations such as SESTM and LMMD fail to generate positive net performance even at the lowest turnover level ($\gamma = 0.1$), whereas LLM-based strategies—with the exception of BERT—continue to deliver economically meaningful Sharpe ratios. For alert-based strategies, LLMs consistently outperform word-based methods across turnover constraints. Notably, Word2Vec achieves relatively strong performance—second only to ChatGPT—suggesting that simpler text representations can still retain a substantial share of the information content in alert-based news.

Table 10 further isolates the role of the downstream predictive model by comparing Lasso, ridge, random forest, and neural networks using a common set of LLaMA3 embeddings. Moving from linear models to neural networks yields substantial improvements in both gross and net performance for both articles and alerts. Random forest and neural networks dominate Lasso and ridge regression across nearly all turnover levels, underscoring the value of flexible predictive models in extracting economic value from rich text representations. Interestingly, ChatGPT combined with ridge regression (as reported in Table 9) can

outperform random forest and neural networks built on LLaMA3 embeddings, suggesting that improvements in text representation quality can, to some extent, substitute for greater flexibility in downstream prediction models.

Table 9: Net Sharpe Ratio Summary Across Language Models

Panel A: Article

γ	ChatGPT	LLaMA3	LLaMA2	LLaMA	RoBERTa	BERT	Word2Vec	SESTM	LMMD
0.10	0.71	0.77	0.68	0.54	0.35	0.06	0.08	-0.00	-0.42
0.20	0.70	0.72	0.64	0.48	0.11	-0.28	-0.34	-0.62	-0.97
0.30	0.58	0.53	0.47	0.29	-0.22	-0.67	-0.82	-1.15	-1.49
0.40	0.42	0.29	0.25	0.05	-0.57	-1.06	-1.29	-1.62	-1.96
0.50	0.23	0.04	0.02	-0.21	-0.90	-1.43	-1.71	-2.02	-2.39
0.60	0.04	-0.21	-0.22	-0.48	-1.21	-1.79	-2.10	-2.37	-2.78
0.70	-0.15	-0.45	-0.44	-0.74	-1.51	-2.12	-2.43	-2.67	-3.12
0.80	-0.33	-0.67	-0.66	-0.99	-1.78	-2.41	-2.73	-2.92	-3.42
0.90	-0.51	-0.87	-0.87	-1.24	-2.02	-2.68	-2.99	-3.13	-3.67

Panel B: Alerts

γ	ChatGPT	LLaMA3	LLaMA2	LLaMA	RoBERTa	BERT	Word2Vec	SESTM	LMMD
0.10	1.48	1.26	1.19	0.91	1.02	0.27	1.54	0.72	0.10
0.20	1.95	1.53	1.51	1.15	1.31	0.20	1.83	0.82	0.08
0.30	2.23	1.65	1.65	1.23	1.36	0.05	1.92	0.80	0.01
0.40	2.40	1.68	1.69	1.22	1.30	-0.11	1.91	0.75	-0.09
0.50	2.49	1.66	1.67	1.16	1.19	-0.28	1.86	0.70	-0.20
0.60	2.51	1.60	1.60	1.07	1.06	-0.42	1.78	0.64	-0.31
0.70	2.49	1.52	1.51	0.97	0.93	-0.56	1.69	0.58	-0.41
0.80	2.44	1.43	1.41	0.86	0.80	-0.67	1.58	0.52	-0.50
0.90	2.35	1.33	1.29	0.74	0.66	-0.78	1.46	0.46	-0.58

Note: This table reports the transaction cost–adjusted performance of LLM-based trading strategies across language models. Results are shown separately for portfolios constructed from full articles (Panel A) and alerts (Panel B). Portfolios are equally weighted long–short portfolios formed using EWCT signals, with the smoothing parameter γ governing portfolio turnover. Sharpe ratios are annualized and computed after accounting for transaction costs. Higher values of γ correspond to higher turnover.

3.6 Polyglot Evidence from International Equity Markets

As a final return-prediction exercise, we evaluate news-based models outside the United States. Because news coverage is sparse in several individual non-U.S. markets (e.g., Portugal, Greece, and the Netherlands), we pool all non-U.S. stocks into a single training set and estimate a unified predictive model.

Our pipeline does not rely on machine translation. Articles are processed in their original languages and encoded by a multilingual language model into a shared cross-lingual embedding space. This common representation allows us to train a single return-prediction model that operates uniformly across languages and countries. In this sense, the same specification applies seamlessly across markets—analogous to technical trading strategies, where

Table 10: Net Sharpe Ratio Summary Across Predictive Models: LLaMA3

	Article				Alert			
γ	Ridge	Lasso	RF	NN	Ridge	Lasso	RF	NN
0.10	0.77	0.33	1.02	0.89	1.26	0.24	1.51	1.44
0.20	0.72	0.19	1.25	1.11	1.53	1.07	1.98	1.83
0.30	0.53	0.06	1.25	1.08	1.65	1.07	2.20	2.00
0.40	0.29	-0.07	1.12	0.95	1.68	1.02	2.27	2.05
0.50	0.04	-0.18	0.95	0.76	1.66	0.94	2.27	2.03
0.60	-0.21	-0.28	0.74	0.55	1.60	0.85	2.22	1.98
0.70	-0.45	-0.36	0.54	0.34	1.52	0.75	2.14	1.90
0.80	-0.67	-0.43	0.33	0.13	1.43	0.65	2.04	1.80
0.90	-0.87	-0.48	0.13	-0.07	1.33	0.55	1.93	1.69

Note: This table compares transaction cost–adjusted performance across predictive models using LLaMA3 embeddings. Portfolios are equally weighted long–short EWCT portfolios constructed from articles and alerts, with turnover governed by the smoothing parameter γ. Sharpe ratios are annualized and reported net of transaction costs.

prices and volumes are directly comparable—without requiring country-specific adjustments to account for heterogeneous accounting standards or disclosure regimes.

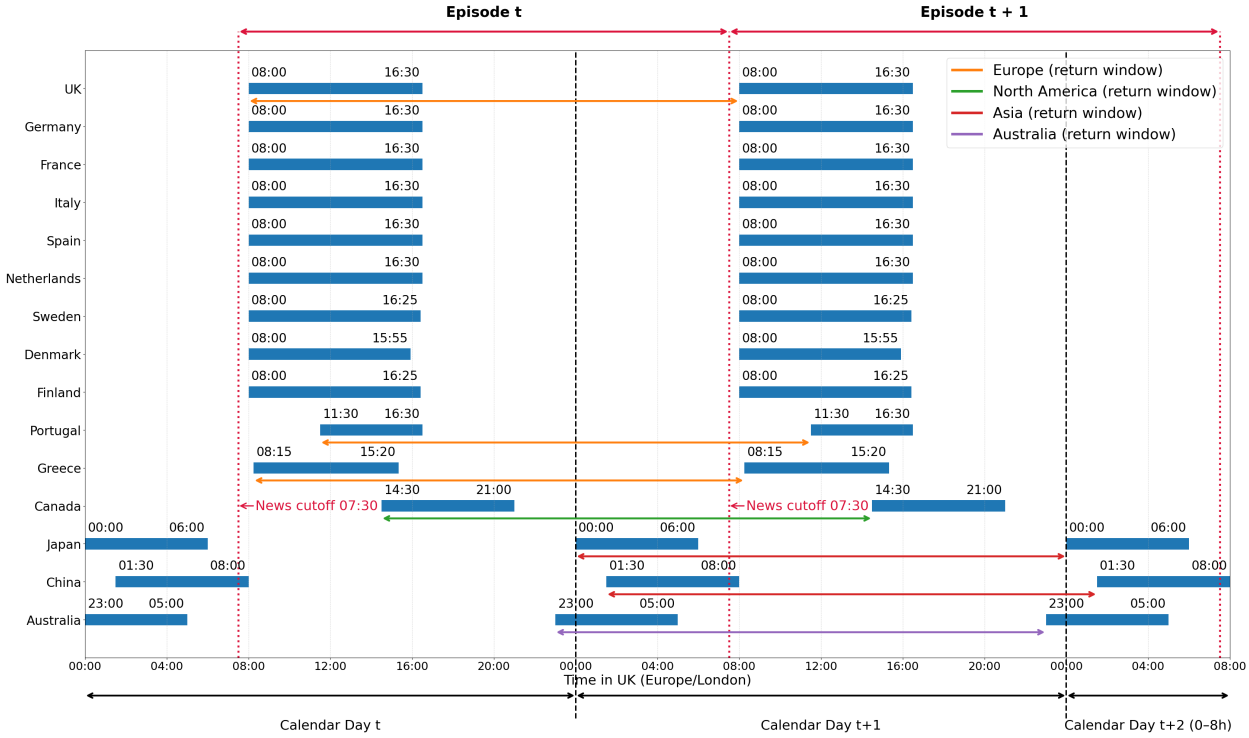
To synchronize news arrival across time zones while respecting local market trading hours, we impose a global news cut-off at 07:30 London time (GMT/BST). This cut-off defines a discrete episode indexed by t, representing an information-aligned period rather than a local calendar day. Return prediction is based on all news observed up to the cut-off, and episode $t+1$ begins at the subsequent 07:30 London cut-off. Asset-level returns are measured using open-to-open price changes, with opening prices adjusted for dividends and stock splits. Let $o(i,t)$ denote the clock time of the local cash market open of asset i associated with episode t, and let $P^{LC}_{i,o(i,t)}$ denote the corresponding local-currency opening price. The local-currency return realized over episode t is given by

$$R^{LC}_{i,t+1} = \left(P^{LC}_{i,o(i,t+1)} - P^{LC}_{i,o(i,t)}\right)/P^{LC}_{i,o(i,t)}.$$

The timing conventions are illustrated in Figure 8. Episode t corresponds to the 24-hour information window ending at the global news cut-off. Returns are measured from the first local cash market open that occurs strictly after the cut-off to the subsequent local open. For markets whose cash session opens after the cut-off on the same local calendar day (e.g., Europe, the U.K., and North America), the relevant opening price is observed on that day. For markets whose cash session has already begun by the cut-off (e.g., China, Japan, and Australia), the relevant opening price is observed at the local open on the following trading day. In all cases, the resulting open-to-open return is attributed to episode t, as it reflects the first tradable price response to information accumulated during that episode.[21]

[21]This timing convention aligns information sets across markets while relying only on prices observable

Figure 8: Polyglot Trading Timeline



Note: This figure illustrates the trading timeline for the polyglot international trading strategy. The global news cut-off is set at 07:30 London time (GMT/BST). Positions are entered and exited at the first local market open strictly after the cut-off, ensuring alignment of information arrival across different time zones.

Table 11: International Market Cross-Sectional Return Correlation

	LLaMA3	LLaMA2	LLaMA	RoBERTa	BERT	Word2Vec
UK	3.08	3.28	3.26	2.48	2.27	0.68
Australia	0.60	0.04	0.15	-0.25	-0.65	0.16
Canada	2.27	2.39	2.47	2.02	1.14	1.52
China	2.09	1.18	1.63	0.54	2.00	1.86
Japan	2.21	2.05	2.19	1.75	1.85	1.09
Germany	1.82	1.98	1.95	0.87	1.51	-0.36
Italy	1.45	1.39	1.46	1.04	0.60	0.63
France	2.03	2.33	2.17	0.98	-0.19	0.92
Sweden	1.79	1.10	1.43	0.74	-0.06	1.39
Denmark	-0.52	0.45	-0.44	0.75	1.68	-0.67
Spain	2.98	3.38	2.24	1.07	1.14	-0.87
Finland	3.46	2.22	1.75	3.42	1.03	-0.07
Portugal	4.48	5.77	4.91	0.60	-1.32	-0.91
Greece	-0.26	-2.31	-1.97	-1.21	3.64	0.02
Netherlands	1.53	0.84	1.45	1.46	2.08	1.28
Mean	1.93	1.74	1.64	1.08	1.11	0.44
English Countries Mean	1.98	1.90	1.96	1.42	0.92	0.79
Excl. English Countries Mean	1.92	1.70	1.56	1.00	1.16	0.36

Note: This table reports the cross-sectional return correlations of international markets for different models. Each entry represents the correlation between daily cross-sectional returns of stocks in a specific country and those in other countries, calculated over the entire sample period. The results are reported in percentage points.

We train a single aggregate return-prediction model using local-currency returns for all non-U.S. stocks, estimated via ridge regressions on news embeddings. Table 11 reports cross-sectional return correlations across international markets for each model, where each entry is the average daily cross-sectional correlation between realized stock returns and model-predicted returns within a given country. U.S. correlations are reproduced for reference. Across markets—except for Denmark and Greece, where correlations appear to have the wrong sign—LLM-based models consistently outperform traditional word-based approaches, with particularly pronounced gains in non-English environments. For example, LLaMA3 exhibits largely stable predictive performance, attaining an average correlation of 1.93% across all countries and 1.92% among non-English markets. In contrast, Word2Vec attains 0.79% on average in English-speaking countries and only 0.36% in non-English markets. Overall, these findings indicate that LLM-based embeddings extract signals that generalize more reliably across languages and institutional settings.

To evaluate economic performance from the perspective of a U.S.-based investor, we form trading portfolios based on the model's return predictions and compute portfolio returns in U.S. dollars, incorporating foreign exchange movements over the same holding interval. Let $c(i)$ denote the country in which asset i is traded, and let $S_{c(i),o(i,t)}$ denote the spot exchange

after the information cut-off. The resulting discrete-time returns abstract from intraday financing flows and are equivalent to self-financing returns under standard frictionless assumptions.

rate (USD per unit of local currency) observed at the same local market open. We use Datastream mid-rates for $S_{c(i),o(i,t)}$. Asset-level USD returns are therefore given by

$$R^{USD}_{i,t+1} = \left(P^{LC}_{i,o(i,t+1)} S_{c(i),o(i,t+1)} - P^{LC}_{i,o(i,t)} S_{c(i),o(i,t)}\right) / \left(P^{LC}_{i,o(i,t)} S_{c(i),o(i,t)}\right).$$

This construction ensures that both equity prices and exchange rates are measured at the same local market opens at entry and exit for each episode, so that equity and currency components are realized over an identical open-to-open holding interval, even though this interval may correspond to different local calendar days across markets.

Portfolio returns for episode t are computed as the weighted average of subsequent asset-level USD returns,

$$R_{t+1} = \sum_i w_{i,t} R^{USD}_{i,t+1},$$

where portfolio weights are assigned at the episode cut-off. Table 12 shows that portfolios constructed using LLM-based signals consistently outperform those based on word-level models. Among the LLMs, LLaMA2 delivers the strongest performance, with a Sharpe ratio of 1.69, followed by LLaMA (1.56), LLaMA3 (1.35), RoBERTa (1.23), and BERT (0.87). In contrast, Word2Vec performs substantially worse, with a Sharpe ratio of only 0.10, indicating no economically meaningful predictive power. Notably, performance in non-U.S. markets is weaker than in the U.S., despite pooling all non-U.S. stocks into a single asset pool. This likely reflects substantially less comprehensive news coverage in polyglot markets in our data source, delays in news dissemination arising from asynchronous trading schedules, and greater exposure to exchange-rate dynamics. Nevertheless, the incremental value of LLMs relative to word-based models remains economically meaningful.

4 Understanding Textual Portfolios

This section provides an economic interpretation of text-based trading strategies by examining how textual information is represented, processed, and transformed into return-predictive signals. We begin by studying forward-looking bias in LLM-based representations and assess its economic relevance for return prediction. We then compare alternative textual inputs—headlines, full articles, and summaries—to identify where predictive content resides. Next, we document systematic news momentum in returns and trace its origin to the speed at which news is incorporated into prices. Finally, we analyze signal attribution through the lens of existing factor models and news topics, clarifying the sources, mechanisms, and robustness of textual portfolios.

Table 12: Portfolio Performance Summary Statistics for Non-US Markets

	LLaMA3						LLaMA2					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.77	0.33	0.43	0.30	0.16	0.13	0.85	0.30	0.54	0.31	0.09	0.21
Std	0.34	0.27	0.32	0.33	0.37	0.35	0.34	0.26	0.32	0.34	0.38	0.36
SR	2.31	1.22	1.35	0.89	0.42	0.37	2.50	1.13	1.69	0.92	0.25	0.58
	LLaMA						RoBERTa					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.78	0.31	0.46	0.25	0.12	0.11	0.82	0.34	0.47	0.21	0.19	0.01
Std	0.31	0.27	0.29	0.33	0.37	0.33	0.39	0.26	0.38	0.32	0.37	0.34
SR	2.53	1.16	1.56	0.76	0.34	0.34	2.08	1.31	1.23	0.67	0.51	0.03
	BERT						Word2Vec					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.66	0.40	0.25	0.24	0.17	0.07	0.54	0.50	0.03	0.18	0.19	-0.01
Std	0.30	0.26	0.28	0.31	0.37	0.33	0.30	0.26	0.29	0.33	0.38	0.36
SR	2.18	1.55	0.87	0.79	0.45	0.20	1.80	1.88	0.10	0.56	0.49	-0.04

Note: This table reports Sharpe ratios of long-short portfolios in international markets, constructed using LLaMA3, LLaMA2, LLaMA, RoBERTa, BERT, and Word2Vec based on next-period return predictions. The analysis includes a combined dataset of non-US countries, with trading aligned to UK market hours.

4.1 Assessing Forward-Looking Bias in LLM-Based Embeddings

A central concern in applying LLMs to asset pricing is look-ahead bias—the possibility that models are exposed during training to information realized after a given news event, and that their predictions reflect memorized outcomes rather than inference based on contemporaneous text. For example, if an investor in 2014 were to query ChatGPT about Tesla's prospects and receive a positive assessment, this response could in principle reflect knowledge of Tesla's subsequent performance embedded in the training data, rather than reasoning based solely on information available at the time.

Recent work provides direct evidence on the economic importance of this concern. For instance, [20] construct chronologically consistent large language models that restrict training data to information available up to each point in time and generate timestamped embeddings. Following our return-prediction framework, they form decile long–short portfolios based on next-day return forecasts. Despite relying exclusively on historically available text and thus eliminating exposure to future information, these models achieve portfolio performance comparable to that of state-of-the-art LLMs that do not impose such restrictions (e.g., LLaMA3). This evidence suggests that, in the context of news-based return prediction, forward-looking bias embedded in LLM representations is economically limited.

A stringent test of the economic relevance of look-ahead bias is to evaluate model performance on data that postdate the training cutoffs of the language models. To this end,

Table 13: Portfolio Performance Before and After Knowledge Cutoff Dates

Panel A: BERT

Period	2004-01-01 to 2022-06-01			2004-01-01 to 2018-10-31			2018-10-31 to 2022-06-01		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.35	-0.02	0.37	0.33	0.03	0.30	0.43	-0.24	0.67
Std	0.21	0.23	0.11	0.20	0.22	0.10	0.25	0.27	0.13
SR	1.64	-0.09	3.36	1.64	0.15	2.90	1.69	-0.89	4.97

Panel B: RoBERTa

Period	2004-01-01 to 2022-06-01			2004-01-01 to 2019-07-31			2019-07-31 to 2022-06-01		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.37	-0.06	0.43	0.35	-0.03	0.38	0.45	-0.24	0.69
Std	0.22	0.23	0.11	0.20	0.22	0.10	0.27	0.30	0.14
SR	1.71	-0.27	3.92	1.73	-0.14	3.72	1.69	-0.80	4.91

Note: This table reports the performance of equal-weighted quintile long-short portfolios, along with their respective long and short legs, constructed using BERT and RoBERTa models. Performance is evaluated over three periods: the full sample (January 1, 2004 to June 1, 2022), the pre-cutoff period (ending at the knowledge cutoff dates—October 2018 for BERT and July 2019 for RoBERTa), and the post-cutoff period (thereafter). Portfolios are formed by sorting the cross-section of stocks based on predicted returns.

we extend the evaluation sample through June 2022, allowing us to examine performance before and after the knowledge cutoff dates of BERT (October 2018) and RoBERTa (July 2019).[22] We split the sample into pre- and post-cutoff subsamples. In the post-cutoff period, any return predictability cannot be attributed to information leakage, providing a clean out-of-sample test.

Table 13 reports the results. Panel A presents BERT and Panel B RoBERTa, with performance shown for the full sample and both subsamples. Both models deliver strong long–short performance in the post-cutoff period, with Sharpe ratios of 4.91 for BERT and 4.97 for RoBERTa—levels that are in fact higher than those in the pre-cutoff subsample. Despite the relatively short post-cutoff window, the economic and statistical significance of these Sharpe ratios suggests that look-ahead bias is unlikely to be a primary driver of the return predictability observed prior to the cutoff dates.

To further address concerns related to look-ahead bias, we apply anonymization techniques that mask firm-specific information in the original news text. Our approach builds on the pattern-based algorithms proposed by [16], which identify official company names and product references using the Google Knowledge Graph, and extends them by incorporating a broader set of masking criteria. In addition to company and product names, we systematically anonymize individual identities, dates, and other salient identifiers. This

[22]The knowledge cutoff dates for LLaMA models vary by version and generally fall in 2023 or later. Because these cutoffs occur after the end of our available sample period, an out-of-sample evaluation for LLaMA-based models is not feasible.

Table 14: Masking Success Rates on Full News Dataset

	Alerts	Articles	Both
# Sample	3,285,772	4,123,043	7,408,815
Success Rate	95.73	86.38	90.53

Note: The table summarizes masking success rates achieved using the enhanced pattern-based anonymization method evaluated on the entire news dataset (alerts and articles) spanning from 1996 to 2019. Success rates reflect the proportion of news items successfully anonymized, as assessed by the LLaMA3.1 8B Instruct model.

comprehensive masking strategy is designed to limit the extent to which LLMs can associate input text with memorized training data, thereby reducing potential sources of bias. We further prompt LLaMA3 Instruct to infer the masked company name from the anonymized text. Details of the anonymization procedure are provided in Internet Appendix C. Table 14 reports the results of applying the pattern-based anonymization procedure to the full dataset from 1996 to 2019. The procedure is highly effective, achieving anonymization rates of 86.38% for articles and 95.73% for alerts, indicating that firm-specific identifiers can be removed for the vast majority of news items.

Based on this procedure, we construct two anonymized datasets. The masked dataset applies anonymization to all news items, regardless of whether LLaMA3 can still infer the underlying firm, while the filtered dataset excludes cases in which anonymization fails—approximately 13% of articles and 4% of alerts—yielding a stricter specification. We then repeat the entire analysis, retraining the models separately using embeddings constructed from each dataset and forming return predictions via ridge regressions.

Portfolios constructed from these predictions exhibit similar performance across specifications. As shown in Table 15, Sharpe ratios remain largely unchanged after anonymization and after excluding potentially identifiable observations, suggesting that return predictability is not driven by firm identifiers in the text. Instead, the models appear to rely on broader informational content conveyed by the news. Notably, for news alerts, performance modestly improves under the filtered specification, consistent with the removal of noise or spurious firm-level cues. Taken together, these findings further alleviate concerns about look-ahead bias and underscore the robustness of LLM-based trading strategies.

4.2 Headline, Body, and AI-Generated Summaries

We next examine how the amount and structure of textual context affect return predictability by comparing three representations of the same news article: the headline, the full body text, and an AI-generated summary.

Table 15: Impact of Anonymization on Portfolio Performance

Panel A: Articles

	LLaMA3						LLaMA3 masked					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.43	-0.07	0.50	0.22	0.09	0.12	0.41	-0.05	0.46	0.21	0.13	0.08
Std	0.20	0.22	0.11	0.19	0.20	0.11	0.21	0.22	0.10	0.19	0.20	0.11
SR	2.08	-0.33	4.59	1.11	0.47	1.10	2.01	-0.21	4.38	1.07	0.62	0.73
	LLaMA3 filtered						LLaMA3 masked + filtered					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.39	-0.04	0.43	0.21	0.14	0.07	0.38	-0.01	0.39	0.21	0.13	0.08
Std	0.20	0.22	0.11	0.20	0.20	0.11	0.21	0.21	0.11	0.20	0.20	0.11
SR	1.92	-0.17	3.99	1.06	0.68	0.66	1.85	-0.06	3.71	1.09	0.65	0.74

Panel B: Alerts

	LLaMA3						LLaMA3 masked					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.50	-0.30	0.80	0.31	0.05	0.26	0.47	-0.27	0.74	0.30	0.05	0.24
Std	0.22	0.25	0.17	0.21	0.22	0.16	0.22	0.24	0.15	0.21	0.21	0.15
SR	2.29	-1.21	4.79	1.52	0.23	1.68	2.16	-1.12	4.96	1.42	0.26	1.63
	LLaMA3 filtered						LLaMA3 masked + filtered					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.44	-0.24	0.68	0.30	0.06	0.23	0.43	-0.21	0.64	0.31	0.09	0.22
Std	0.22	0.25	0.16	0.22	0.22	0.15	0.22	0.24	0.14	0.21	0.23	0.16
SR	1.97	-0.96	4.19	1.33	0.29	1.50	1.99	-0.88	4.47	1.46	0.39	1.38

Note: This table presents the impact of anonymization on portfolio performance. Panel A reports the results for news articles, while Panel B presents the results for news alerts. Portfolios are constructed as equal-weighted and value-weighted long, short, and long-short portfolios. We compare four anonymization settings based on LLaMA3 predictions: i) the original model output ; ii) the masked dataset (LLaMA3 masked), where all news are anonymized using the pattern-based procedure regardless of inference success; iii) the filtered dataset (LLaMA3 filtered), which excludes cases where anonymization fails (approximately 13% of articles and 4% of alerts); and iv) the masked filtered dataset (LLaMA3 masked + filtered), which first applies masking and then removes the failed cases.

Headlines provide a concise, editor-curated signal; body text contains richer narrative detail and contextual nuance; and AI-generated summaries offer a compressed representation of the article's content, constructed from article bodies using LLaMA3 Instruct. Comparing headlines with AI-generated summaries allows us to contrast human- and machine-curated forms of information compression, which may differ systematically in structure and emphasis and, consequently, in their effectiveness for downstream return prediction.

For each textual representation—headline, body, and AI-generated summary—we re-estimate the return prediction model using ridge regression on the corresponding embeddings and construct long–short portfolios based on the resulting predictions.

Table 16 reports the performance of long–short portfolios constructed from return pre-

Table 16: Portfolio Performance by News Text Representation

	ChatGPT						LLaMA3					
	Headline		Article		Summary		Headline		Article		Summary	
	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW
Ret	0.50	0.17	0.53	0.16	0.56	0.18	0.34	0.11	0.50	0.12	0.59	0.18
Std	0.11	0.12	0.11	0.12	0.11	0.12	0.09	0.10	0.11	0.11	0.11	0.11
SR	4.57	1.40	4.78	1.34	5.01	1.56	3.59	1.08	4.59	1.10	5.42	1.64
	LLaMA2						LLaMA					
	Headline		Article		Summary		Headline		Article		Summary	
	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW
Ret	0.43	0.13	0.50	0.11	0.61	0.20	0.38	0.11	0.45	0.10	0.60	0.19
Std	0.11	0.11	0.11	0.11	0.11	0.12	0.10	0.11	0.11	0.11	0.11	0.11
SR	4.04	1.21	4.36	1.02	5.63	1.74	3.78	1.00	4.04	0.89	5.69	1.77
	RoBERTa						BERT					
	Headline		Article		Summary		Headline		Article		Summary	
	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW
Ret	0.34	0.09	0.38	0.10	0.52	0.18	0.39	0.07	0.31	0.08	0.47	0.18
Std	0.10	0.11	0.10	0.11	0.10	0.11	0.10	0.11	0.10	0.11	0.10	0.11
SR	3.34	0.82	3.69	0.92	5.04	1.69	3.86	0.70	3.00	0.70	4.52	1.60
	Word2Vec						SESTM					
	Headline		Article		Summary		Headline		Article		Summary	
	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW
Ret	0.40	0.12	0.29	0.07	0.45	0.17	0.58	0.21	0.27	0.09	0.55	0.16
Std	0.10	0.11	0.10	0.11	0.10	0.11	0.15	0.15	0.11	0.12	0.11	0.12
SR	3.92	1.11	2.74	0.60	4.42	1.60	3.90	1.39	2.57	0.74	4.80	1.30

Note: This table presents the performance of equal-weighted and value-weighted long-short portfolios constructed using different text types (article headline, article body, and LLM-generated summary). The portfolios are built based on next-period return predictions generated by various models. All results are annualized.

dictions based on different textual representations. A clear pattern emerges for modern LLMs. For ChatGPT, LLaMA3, LLaMA2, LLaMA, and RoBERTa, portfolios constructed from full article text generally outperform those based on headlines, particularly in equal-weighted specifications. This pattern suggests that richer textual context improves return predictability when the model is capable of processing long-range dependencies and nuanced semantic structure. BERT is a partial exception, as headline-based portfolios sometimes outperform body-text portfolios, indicating a more limited ability to exploit additional textual complexity.

Across LLMs, AI-generated summaries deliver the strongest and most consistent performance. In most cases, summary-based portfolios dominate both headline- and body-based portfolios. For example, under LLaMA3, the equal-weighted Sharpe ratio rises from 3.59 for headlines and 4.59 for full articles to 5.42 for summaries, with a similar ranking in value-weighted portfolios. A comparable ordering holds for LLaMA2, LLaMA, and RoBERTa. These results suggest that summaries retain the most economically relevant information while presenting it in a condensed and structured form that aligns particularly well with LLM embeddings.

For traditional word-based models, the pattern differs. Word2Vec and SESTM do not benefit from longer article text; headline-based portfolios often outperform those using full articles, and performance tends to deteriorate as document length increases. This is consistent with the view that these models struggle to extract incremental signal from richer textual content and may instead accumulate noise. Interestingly, summaries improve performance even for these non-LLM models relative to full articles. However, this gain likely reflects the advantage of shorter, more concentrated text—reducing noise and dimensionality—rather than an enhanced ability to process semantic complexity.

Overall, the evidence indicates that increasing textual richness enhances return predictability only when paired with sufficiently expressive language models. Summaries appear to strike an effective balance by concentrating salient information while reducing extraneous detail. This compression benefits all models, but LLM-based embeddings translate the structured and distilled content of summaries into particularly strong predictive performance.

4.3 News Assimilation and News Momentum

Having established how different forms of textual information generate return predictability, we now study how such information is incorporated into prices at short horizons in the cross section. A robust empirical regularity in this setting is short-term reversal, whereby stocks with high recent returns subsequently underperform, and stocks with low recent returns subsequently outperform, over horizons of days to weeks. A large literature attributes this pattern to temporary price pressure and liquidity provision, rather than to the incorporation of fundamental information (e.g., [22, 28, 8]). In these models, non-informational order flow induces transitory price movements that subsequently revert as liquidity providers unwind inventories. Under this benchmark, return continuation is not expected at short horizons when trading is dominated by liquidity demand rather than information.

We begin by examining whether this benchmark holds uniformly in the presence of firm-specific news. Table 17 reports the performance of long-short portfolios sorted on past five-day cumulative returns, constructed both over the full stock universe and over the subsample of stocks associated with recent news coverage. Consistent with prior evidence, short-term reversal is pronounced in the full universe, with negative long–short returns and Sharpe ratios of -2.22 for equal-weighted portfolios and -1.06 for value-weighted portfolios. When attention is restricted to stocks with news, reversal remains present but is noticeably attenuated: Sharpe ratios weaken to -1.48 and -0.83 in the equal- and value-weighted cases, respectively. While the magnitude of this attenuation is modest, it suggests that the arrival of news partially offsets liquidity-driven price pressure, altering short-horizon cross-sectional

return dynamics.

Table 17: Short-term Reversal in the Cross-Section of Stock Returns

	Entire Market						Stocks with News					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	-0.33	-0.02	-0.30	-0.28	-0.10	-0.19	-0.33	-0.06	-0.26	-0.27	-0.08	-0.19
Std	-0.23	-0.18	0.14	-0.26	-0.20	0.18	-0.27	-0.22	0.18	-0.29	-0.22	0.23
SR	-1.44	-0.12	-2.22	-1.07	-0.47	-1.06	-1.21	-0.27	-1.48	-0.95	-0.37	-0.83

Note: This table reports the performance of long–short portfolios sorted on past five-day cumulative returns. Portfolios are constructed using equal-weighted and value-weighted schemes. The left panel uses the universe of all CRSP common stocks (NYSE, AMEX, NASDAQ), while the right panel restricts the investment universe to stocks associated with recent news coverage. Returns and standard deviations are reported in daily percentage terms; Sharpe ratios are annualized.

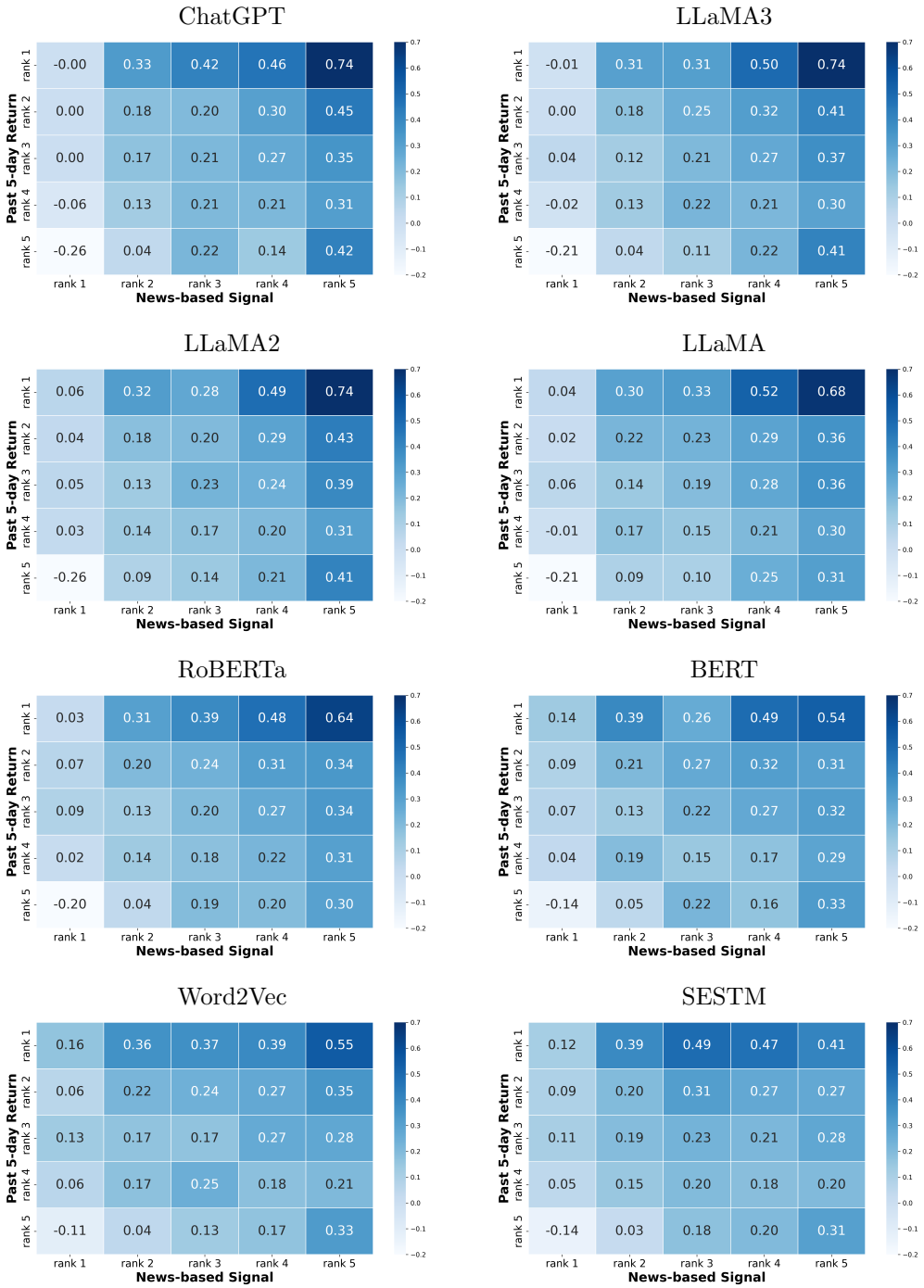
To distinguish more sharply between liquidity-driven effects and information-driven predictability, we next examine whether news-based signals provide incremental information beyond past returns. We conduct a double-sorting analysis in which stocks are independently sorted on past five-day returns and on news-based return predictions, forming 5×5 portfolios for each predictive model. Figure 9 reveals two clear and systematic patterns. Along the past-return dimension, average portfolio returns increase monotonically from bottom to top, consistent with short-term reversal. Along the news-signal dimension, returns increase from left to right within each past-return quintile, indicating that news-based signals contain predictive information that is incremental to past returns.

Across all past-return bins, portfolios formed using LLM-based signals deliver higher returns than those based on word-level representations, indicating superior informational content conditional on past returns. Taken together, these results show that contextual news information adds predictive value beyond traditional reversal signals and is not simply a transformation of recent price movements.

The role of news becomes even more pronounced when portfolios are formed directly on news-based return predictions. Table 18 reports the performance of long–short portfolios constructed using next-period return forecasts and traded within the universe of news-tagged stocks. We observe a clear and economically significant short-term momentum pattern: returns are strongly positive on the first trading day following portfolio formation and then decay monotonically over subsequent days. For equal-weighted portfolios, Sharpe ratios exceed four on Day 1 before declining steadily. This behavior contrasts sharply with the baseline reversal pattern and indicates that when trading is conditioned on news-based information, prices exhibit short-run continuation rather than mean reversion.

These findings support an interpretation of momentum as arising from gradual information assimilation. The strong initial return response followed by monotonic decay suggests

Figure 9: Double-Sorted Portfolio Returns



Note: This figure presents double-sorted long-short equal weighted portfolio returns across different model types, based on next-period return prediction signals. Portfolios are sorted first on 5-day past returns and then on news-based signals. Each panel corresponds to a distinct model. Rank 1 indicates the lowest values for past returns or news-based signals.

Table 18: Delayed Portfolio Performance

	Day+1 Portfolio						Day+2 Portfolio					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.43	-0.07	0.50	0.22	0.09	0.12	0.18	0.05	0.13	0.08	0.10	-0.02
Std	0.20	0.22	0.11	0.19	0.20	0.11	0.20	0.21	0.10	0.19	0.19	0.11
SR	2.08	-0.33	4.58	1.11	0.47	1.10	0.92	0.24	1.35	0.40	0.52	-0.22
	Day+3 Portfolio						Day+4 Portfolio					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.17	0.08	0.09	0.09	0.09	-0.01	0.17	0.09	0.08	0.11	0.08	0.03
Std	0.20	0.21	0.09	0.19	0.19	0.11	0.20	0.21	0.09	0.20	0.19	0.11
SR	0.86	0.39	0.98	0.46	0.47	-0.05	0.86	0.45	0.91	0.56	0.40	0.31
	Day+5 Portfolio						Day+6 Portfolio					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.19	0.14	0.05	0.09	0.06	0.03	0.14	0.12	0.02	0.15	0.08	0.08
Std	0.20	0.21	0.09	0.20	0.19	0.11	0.20	0.21	0.09	0.20	0.20	0.11
SR	0.93	0.67	0.54	0.48	0.33	0.27	0.70	0.56	0.25	0.76	0.39	0.73

Note: This table reports the LLaMA3's performance of equal-weighted and value-weighted long-short portfolios sorted on next-period return predictions and their long and short legs in six consecutive days.

that markets react quickly to news, but that full price adjustment unfolds over a short sequence of trading days rather than instantaneously. Momentum therefore reflects delayed, yet directionally correct, incorporation of information, rather than price overshooting and subsequent correction.

To directly assess the speed of this assimilation process, we exploit the sequential structure of news dissemination. Using a database of chronologically ordered news alerts within developing stories, we train models on the full set of alerts and construct portfolios based on different subsets of alerts. Two patterns emerge in Table 19. First, portfolios based on news alerts consistently generate higher Sharpe ratios than those based on full news articles, with the exception of BERT. Despite their limited textual content, alerts convey highly concentrated information in a timely manner, underscoring the importance of speed over richness for short-horizon trading. Consistent with this interpretation, the performance gap between LLMs and word-based models narrows in the alert-based setting, indicating that timeliness alone allows even simpler models to extract economically meaningful signals.

Second, Sharpe ratios decline sharply as we move from the initial alert in a news sequence (TS1) to subsequent alerts (TS2) and later follow-ups. By the time a first update in a developing story is released, prices have largely adjusted, leaving little incremental information to exploit. This pattern is consistent with rapid assimilation of headline information upon

Table 19: Sharpe Ratios of Portfolios Based on News Articles and News Alerts

	EW					VW				
	Article	Alert				Article	Alert			
		All	TS1	TS2	Rest		All	TS1	TS2	Rest
ChatGPT	4.78	5.81	4.13	2.78	2.27	1.34	2.25	1.80	0.53	0.60
LLaMA3	4.59	4.79	3.79	3.02	2.72	1.10	1.68	1.54	0.83	0.84
LLaMA2	4.36	4.67	3.59	2.72	2.55	1.02	1.63	1.07	0.92	0.56
LLaMA	4.04	4.23	3.58	2.94	2.13	0.89	1.56	1.25	1.03	0.57
RoBERTa	3.69	4.34	3.94	3.00	2.57	0.92	1.74	1.55	0.84	0.89
BERT	3.00	2.95	3.10	2.67	2.13	0.70	1.00	1.68	0.92	0.96
Word2Vec	2.74	5.28	4.62	3.32	2.66	0.60	1.77	1.79	1.04	1.21
SESTM	2.57	2.87	2.04	1.75	2.07	0.74	1.12	1.19	0.43	0.32
LMMD	2.33	2.95	2.78	0.95	1.01	0.40	1.16	1.07	0.15	0.25

Note: This table reports Sharpe Ratios for long-short quintile portfolios formed on the basis of ChatGPT, LLaMA3, LLaMA2, LLaMA, RoBERTa, BERT, Word2Vec, SESTM, and LMMD models, sorted by expected return predictions. "TS1" denotes portfolios relying exclusively on the first sequence of alerts, "TS2" uses only the second sequence, "Rest" encompasses all remaining alerts, and "All" includes every alert.

first release.

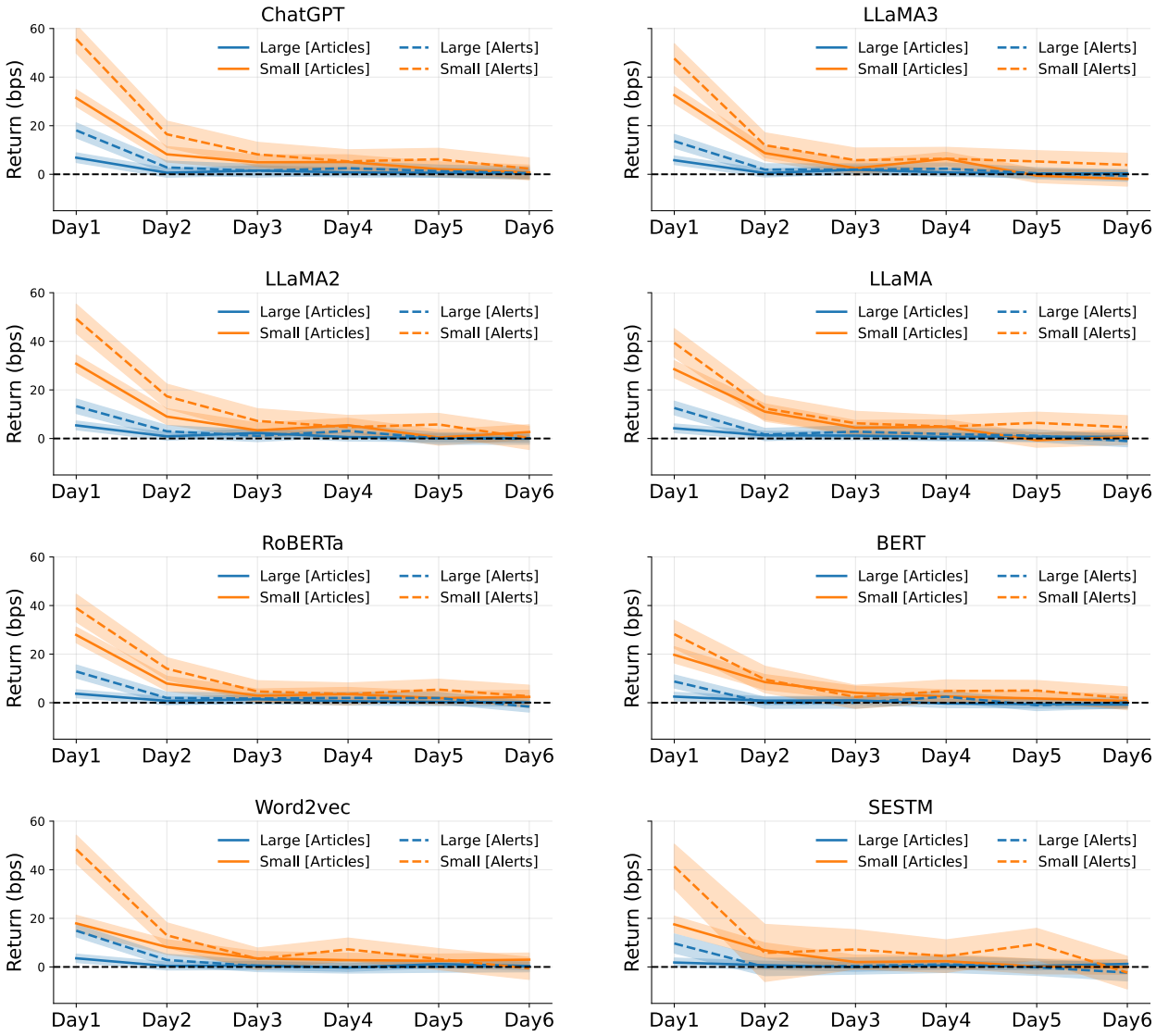
We further quantify assimilation speed by delaying portfolio formation by one to six trading days following news arrival and by examining heterogeneity across firm size. As shown in Figure 10, both article- and alert-based portfolios generate strong immediate return responses, particularly among small stocks. For large firms, return predictability dissipates quickly, with most effects absorbed within one to two trading days. In contrast, for small firms, economically meaningful return responses persist for three to four days following news arrival across both articles and alerts, especially for LLM-based models. This size-dependent pattern indicates that while markets react promptly to news, full price adjustment occurs more slowly for smaller firms, consistent with greater information frictions and delayed interpretation at the firm level.

Taken together, these results provide a unified view of news-driven return dynamics. News weakens short-term reversal and induces transient momentum because information is incorporated rapidly yet sequentially. Alerts generate sharp, short-lived price responses, while articles produce slower but more persistent effects. Short-term momentum emerges in the narrow window between news arrival and full assimilation, highlighting the central role of timing, sequencing, and contextual understanding in news-based trading strategies.

4.4 Which Drives Return Predictability?

An important question is whether the observed returns reflect compensation for exposure to known risk factors or represent genuine abnormal performance. To address this, we estimate factor-adjusted alphas for the LLM-based long–short portfolios using time-series regressions on standard expected-return benchmarks. We consider the hedged DMRS factors of [9],

Figure 10: News Assimilation Across Firm Size



Note: This figure illustrates the speed of news assimilation across models and firm size categories. Each panel corresponds to a specific model, with article-based and alert-based strategies plotted together for comparison. We focus on equal-weighted portfolios. Returns are average one-day holding-period (open-to-open) returns from a news-based trading strategy, evaluated as a function of the delay between news arrival and portfolio formation. Trades are initiated one to six trading days after the announcement. Reported returns are in basis points per day, with shaded regions denoting 95% confidence intervals. Results are shown for ChatGPT, LLaMA3, LLaMA2, LLaMA, RoBERTa, BERT, Word2Vec, SESTM, and LMMD.

the q-factors of [21], and momentum, short-term reversal, and long-term reversal from Ken French's data library. Our trading rule remains unchanged: portfolios are formed at the market open and held open-to-open. Because the standard factor series are reported close-to-close, we convert our portfolio returns using CRSP open and close prices together with CRSP adjusted close-to-close returns. This procedure effectively values each day's portfolio at the close while preserving the underlying open-to-open trading strategy. Tables 20 and 21 report annualized alphas and test statistics. Across models and for both equal-weighted and value-weighted portfolios, the intercepts are positive and economically large, and in most cases are close to the corresponding average portfolio returns, indicating that standard factor exposures explain only a limited fraction of the strategy's performance.

We further broaden the analysis by benchmarking against the JKP factor portfolios of [24], which comprise 153 characteristic-based factors grouped into 13 thematic clusters. Since the JKP factors are constructed at a monthly frequency, we form a monthly analogue of our daily strategies using the turnover-controlled portfolio framework in Section 3.5. Specifically, we focus on the smoothing parameter $\gamma = 0.1$, under which portfolio weights adjust gradually so that a full rebalance occurs over roughly one month, and we aggregate daily portfolio returns to the monthly horizon. We then regress monthly portfolio returns on the 13 thematic JKP factors, with results reported in Tables 22 and 23. The equal-weighted portfolio earns an average annual return of approximately 10%. While the 13 thematic JKP factors absorb a nontrivial share of the return variation, the regression intercept remains economically sizable at roughly 6% per year. This alpha remains positive and economically meaningful. In contrast, the value-weighted alphas are not statistically significant. Overall, these results suggest that the strategy's performance is not primarily driven by exposure to broad characteristic-based factor structure, and that the LLM-based signals contain incremental information beyond existing benchmarks, with the strongest abnormal performance concentrated among smaller stocks.

4.5 Topic Decomposition of News Signals

To better understand the economic mechanisms underlying the documented predictability, we examine which types of information drive the forecasts. Specifically, we analyze the latent thematic structure of the traded news and study cross-sectional profitability across topics.

We segment the news corpus into semantically coherent categories. Because raw text is unstructured, organizing it into themes typically requires a topic model. Here, we provide a convenient alternative that does not rely on a structural modeling framework. LLM embeddings map textual content into a dense semantic space that preserves rich syntactic and

Table 20: Alpha Test Performance: All Factors (EW)

	ChatGPT	LLaMA3	LLaMA2	LLaMA	RoBERTa	BERT	SESTM	Word2Vec	LMMD
MktRF_DMRS	−0.011	−0.061**	−0.103***	−0.090***	−0.015	−0.129***	−0.030	−0.061**	0.032
	(0.030)	(0.029)	(0.030)	(0.030)	(0.028)	(0.027)	(0.124)	(0.028)	(0.025)
HML_DMRS	−0.233***	−0.135***	−0.114**	−0.116**	−0.128***	−0.105**	0.192	−0.067	−0.179***
	(0.045)	(0.044)	(0.046)	(0.046)	(0.042)	(0.041)	(0.186)	(0.042)	(0.038)
RMW_DMRS	−0.045	−0.015	−0.039	0.008	−0.020	−0.008	0.103	−0.080	−0.020
	(0.060)	(0.059)	(0.062)	(0.061)	(0.057)	(0.056)	(0.251)	(0.057)	(0.052)
CMA_DMRS	0.023	0.034	0.088	−0.010	0.045	0.020	0.180	−0.021	0.090
	(0.068)	(0.066)	(0.070)	(0.070)	(0.064)	(0.063)	(0.284)	(0.065)	(0.059)
SMB_DMRS	0.146***	0.141***	0.150***	0.180***	0.119**	0.133***	−0.052	0.091*	0.014
	(0.051)	(0.050)	(0.053)	(0.052)	(0.048)	(0.048)	(0.214)	(0.049)	(0.044)
R_MKT	0.031	0.104***	0.136***	0.119***	0.052***	0.125***	0.044	0.049**	−0.020
	(0.021)	(0.021)	(0.022)	(0.022)	(0.020)	(0.020)	(0.089)	(0.020)	(0.018)
R_ME	0.009	−0.016	−0.061	−0.100**	−0.022	−0.059	0.077	−0.072*	0.038
	(0.039)	(0.038)	(0.040)	(0.040)	(0.037)	(0.036)	(0.163)	(0.037)	(0.034)
R_IA	0.137***	0.009	−0.020	0.011	0.075	0.144***	−0.112	0.016	−0.011
	(0.050)	(0.049)	(0.052)	(0.051)	(0.047)	(0.047)	(0.209)	(0.048)	(0.043)
R_ROE	0.358***	0.332***	0.428***	0.359***	0.308***	0.289***	0.135	0.308***	0.138***
	(0.038)	(0.038)	(0.040)	(0.039)	(0.036)	(0.036)	(0.161)	(0.037)	(0.033)
R_EG	0.042	0.007	−0.014	−0.031	0.026	0.044	0.112	−0.014	0.038
	(0.040)	(0.039)	(0.041)	(0.041)	(0.038)	(0.037)	(0.167)	(0.038)	(0.034)
LT_Rev	−0.020	−0.027	−0.041	−0.004	0.046*	0.007	−0.064	0.068**	0.037
	(0.029)	(0.028)	(0.030)	(0.029)	(0.027)	(0.027)	(0.120)	(0.027)	(0.025)
ST_Rev	−0.053***	−0.043***	−0.047***	−0.064***	−0.028**	−0.056***	−0.022	−0.045***	−0.019
	(0.015)	(0.014)	(0.015)	(0.015)	(0.014)	(0.014)	(0.062)	(0.014)	(0.013)
Mom	0.058***	0.015	0.034**	0.048***	0.026*	0.055***	0.096	0.025*	0.104***
	(0.016)	(0.016)	(0.017)	(0.016)	(0.015)	(0.015)	(0.067)	(0.015)	(0.014)
Constant	0.516***	0.485***	0.483***	0.443***	0.375***	0.309***	0.192*	0.287***	0.216***
	(0.028)	(0.027)	(0.029)	(0.028)	(0.026)	(0.026)	(0.116)	(0.026)	(0.024)
Average Ret	0.529	0.497	0.496	0.451	0.386	0.315	0.287	0.208	0.225
Adj R-squared	0.100	0.049	0.063	0.054	0.049	0.055	−0.001	0.036	0.082

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table reports the results of regressing equal-weighted portfolio returns on an augmented set of factors that includes HXZ, DMRS and additional factors. The portfolios are built on the basis of ChatGPT, LLaMA3, LLaMA2, LLaMA, RoBERTa, BERT, SESTM, Word2Vec and LMMD, respectively. All portfolio returns and factors are annualized.

contextual information. Nevertheless, the high dimensionality of these embeddings (e.g., 4,096 dimensions) can create computational and statistical challenges for clustering. We therefore adopt a two-step projection procedure. First, each article is transformed into an embedding using a pre-trained LLM. Second, we apply dimensionality reduction to obtain a low-dimensional representation suitable for clustering.

Specifically, we employ t-SNE [35] to project the embeddings into a two-dimensional manifold that preserves local similarity, followed by K-means clustering to identify semantic groups. We restrict attention to articles associated with stocks entering our daily long–short

Table 21: Alpha Test Performance on Daily Portfolio: All Factors (VW)

	ChatGPT	LLaMA3	LLaMA2	LLaMA	RoBERTa	BERT	SESTM	Word2Vec	LMMD
MktRF_DMRS	−0.058*	−0.068**	−0.061**	−0.032	−0.026	−0.077***	0.005	−0.106***	0.086***
	(0.030)	(0.029)	(0.030)	(0.029)	(0.028)	(0.029)	(0.124)	(0.029)	(0.026)
HML_DMRS	−0.548***	−0.299***	−0.231***	−0.377***	−0.331***	−0.350***	−0.047	−0.258***	−0.369***
	(0.045)	(0.044)	(0.045)	(0.044)	(0.042)	(0.044)	(0.187)	(0.044)	(0.040)
RMW_DMRS	−0.093	−0.106*	−0.182***	0.060	0.006	−0.080	0.092	−0.208***	−0.016
	(0.061)	(0.060)	(0.060)	(0.059)	(0.057)	(0.059)	(0.252)	(0.059)	(0.054)
CMA_DMRS	0.207***	0.060	0.070	0.119*	0.187***	0.407***	0.396	0.261***	0.247***
	(0.069)	(0.068)	(0.068)	(0.067)	(0.065)	(0.066)	(0.285)	(0.066)	(0.061)
SMB_DMRS	0.015	−0.057	−0.043	−0.007	−0.050	0.008	−0.001	−0.108**	−0.040
	(0.052)	(0.051)	(0.051)	(0.050)	(0.049)	(0.050)	(0.215)	(0.050)	(0.046)
R_MKT	−0.012	−0.001	0.025	0.003	−0.032	0.016	−0.012	0.035*	−0.113***
	(0.022)	(0.021)	(0.021)	(0.021)	(0.020)	(0.021)	(0.090)	(0.021)	(0.019)
R_ME	0.094**	0.145***	0.107***	0.058	0.057	0.010	0.028	0.073*	0.093***
	(0.040)	(0.039)	(0.039)	(0.038)	(0.037)	(0.038)	(0.164)	(0.038)	(0.035)
R_IA	0.129**	0.046	−0.052	0.020	0.151***	0.186***	0.021	0.064	0.168***
	(0.051)	(0.050)	(0.050)	(0.049)	(0.048)	(0.049)	(0.210)	(0.049)	(0.045)
R_ROE	0.008	0.101***	0.152***	−0.050	0.020	0.099***	−0.042	0.073*	−0.060*
	(0.039)	(0.038)	(0.038)	(0.038)	(0.037)	(0.038)	(0.161)	(0.038)	(0.034)
R_EG	0.023	−0.043	−0.053	0.058	0.036	−0.115***	−0.001	0.012	0.044
	(0.041)	(0.040)	(0.040)	(0.039)	(0.038)	(0.039)	(0.168)	(0.039)	(0.036)
LT_Rev	−0.095***	−0.034	−0.036	−0.024	0.027	−0.054*	−0.293**	−0.093***	0.012
	(0.029)	(0.029)	(0.029)	(0.028)	(0.027)	(0.028)	(0.121)	(0.028)	(0.026)
ST_Rev	−0.010	0.010	−0.004	0.040***	−0.030**	−0.020	0.004	−0.052***	−0.028**
	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)	(0.014)	(0.062)	(0.014)	(0.013)
Mom	0.095***	0.019	−0.004	0.037**	0.046***	0.054***	0.111*	0.057***	0.057***
	(0.016)	(0.016)	(0.016)	(0.016)	(0.015)	(0.016)	(0.067)	(0.016)	(0.014)
Constant	0.156***	0.124***	0.114***	0.088***	0.102***	0.085***	0.010	0.070***	0.041*
	(0.028)	(0.028)	(0.028)	(0.027)	(0.026)	(0.027)	(0.116)	(0.027)	(0.025)
Average Ret	0.162	0.125	0.114	0.100	0.102	0.080	0.067	0.025	0.043
Adj R-squared	0.107	0.035	0.022	0.044	0.057	0.061	0.002	0.042	0.096

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Same as Table 20, but for value-weighted portfolios.

portfolio. Each day, we form a zero-net-investment portfolio by going long the top quintile of predicted returns and short the bottom quintile, and clustering is performed only on news linked to these traded stocks. To improve semantic purity, we remove boilerplate content prior to embedding by using article summaries. Typical examples include instructions such as "Double click on the URL below to view the article," copyright notices, and author metadata. Using summaries materially improves the quality of the embedding space.

To uncover higher-level thematic structure, we follow [7] and apply agglomerative clustering to the centroids of the initial 200 K-means clusters. This hierarchical merging procedure balances thematic granularity and interpretability and yields 10 economically meaningful

Table 22: Alpha Test Performance: JKP Factors (EW)

	ChatGPT	LLaMA3	LLaMA2	LLaMA	RoBERTa	BERT	SESTM	Word2Vec	LMMD
Accruals	0.199	0.280	0.177	0.244	0.215	0.035	−0.021	0.213	−0.181
	(0.302)	(0.290)	(0.266)	(0.288)	(0.242)	(0.255)	(0.157)	(0.216)	(0.121)
Debt Issuance	0.184	0.105	0.270	0.101	−0.058	−0.105	−0.504**	0.065	0.075
	(0.323)	(0.333)	(0.394)	(0.392)	(0.277)	(0.285)	(0.237)	(0.278)	(0.209)
Investment	0.326	0.161	0.192	0.296	0.330*	0.351*	0.162	0.026	0.077
	(0.199)	(0.271)	(0.252)	(0.252)	(0.179)	(0.204)	(0.157)	(0.195)	(0.171)
Low Leverage	−0.423	−0.756***	−0.927***	−0.745***	−0.657***	−0.745***	−0.194	−0.747***	−0.277*
	(0.310)	(0.238)	(0.205)	(0.265)	(0.195)	(0.197)	(0.180)	(0.182)	(0.142)
Low Risk	0.202**	0.087	0.125	0.120	0.082	0.022	0.087*	0.128**	0.136**
	(0.086)	(0.076)	(0.090)	(0.081)	(0.076)	(0.071)	(0.050)	(0.055)	(0.057)
Momentum	−0.165	−0.141*	−0.204**	−0.188*	−0.111	−0.134*	−0.004	−0.123**	−0.039
	(0.117)	(0.084)	(0.099)	(0.103)	(0.083)	(0.080)	(0.058)	(0.059)	(0.055)
Profit Growth	0.397*	0.225	0.167	0.329	0.028	0.095	0.182	−0.021	0.222**
	(0.232)	(0.204)	(0.168)	(0.225)	(0.157)	(0.211)	(0.146)	(0.168)	(0.108)
Profitability	0.189	−0.035	0.277	0.173	0.259	0.210	−0.193	0.178	−0.140
	(0.225)	(0.235)	(0.221)	(0.214)	(0.171)	(0.196)	(0.142)	(0.186)	(0.182)
Quality	0.741***	0.873***	0.873***	0.827***	0.624***	0.652***	0.473***	0.476***	0.467***
	(0.161)	(0.162)	(0.134)	(0.159)	(0.112)	(0.127)	(0.099)	(0.123)	(0.081)
Seasonality	−0.318	−0.257	−0.156	−0.173	−0.148	−0.127	−0.073	0.083	−0.107
	(0.335)	(0.254)	(0.282)	(0.327)	(0.187)	(0.215)	(0.140)	(0.168)	(0.137)
Short-Term Reversal	−0.049	0.025	0.034	−0.096	0.013	0.093	−0.032	0.158*	0.083
	(0.122)	(0.085)	(0.092)	(0.112)	(0.092)	(0.125)	(0.103)	(0.093)	(0.103)
Size	0.215	0.236	0.376**	0.319*	0.328**	0.243*	0.051	0.248*	0.060
	(0.166)	(0.157)	(0.179)	(0.169)	(0.127)	(0.134)	(0.078)	(0.128)	(0.096)
Value	−0.551**	−0.653***	−0.862***	−0.699***	−0.662***	−0.711***	−0.091	−0.580***	−0.368***
	(0.266)	(0.213)	(0.234)	(0.256)	(0.221)	(0.218)	(0.173)	(0.166)	(0.126)
Constant	0.056***	0.064***	0.059***	0.049***	0.047***	0.040***	0.040***	0.037***	0.021***
	(0.017)	(0.013)	(0.012)	(0.014)	(0.011)	(0.011)	(0.010)	(0.010)	(0.008)
Average Ret.	0.100	0.094	0.094	0.086	0.072	0.059	0.058	0.055	0.036
Adj R-squared	0.389	0.337	0.389	0.327	0.296	0.269	0.339	0.323	0.483

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Same as Table 20, except for the use of the 13 thematic JKP factors at the monthly frequency. All portfolio returns and factor returns are annualized.

meta-topics.[23]

The ten meta-topics obtained from the hierarchical clustering, ranked by their document frequencies, are as follows: Operating Performance & Earnings Signals (blue), Business Conditions & Economic Developments (purple), Operational & Technology Initiatives (green), Corporate Transactions & Strategic Actions (cyan), Corporate Investment & Fi-

[23]The exact number of initial K-means clusters and the final number of meta-topics has little impact on the results. Agglomerative clustering produces a nested hierarchy of partitions, so alternative cut levels of the dendrogram correspond to coarser or finer groupings of largely similar underlying themes. Varying the initial cluster count between 150 and 300, and selecting between 8 and 15 meta-topics, yields economically similar topic compositions and cross-sectional return patterns.

Table 23: Alpha Test Performance: JKP Factors (VW)

	ChatGPT	LLaMA3	LLaMA2	LLaMA	RoBERTa	BERT	SESTM	Word2Vec	LMMD
Accruals	0.467*	0.259	0.398*	0.567**	0.293	0.352	0.198	0.192	0.318**
	(0.257)	(0.217)	(0.226)	(0.223)	(0.225)	(0.275)	(0.207)	(0.183)	(0.145)
Debt Issuance	1.292**	1.066**	1.422***	1.440***	0.811*	1.020**	0.143	0.613*	0.643**
	(0.592)	(0.442)	(0.519)	(0.478)	(0.431)	(0.462)	(0.385)	(0.330)	(0.262)
Investment	−0.532	−0.626***	−0.878***	−0.759***	−0.292	−0.309	−0.176	−0.344	−0.072
	(0.385)	(0.221)	(0.302)	(0.273)	(0.257)	(0.299)	(0.324)	(0.234)	(0.158)
Low Leverage	−0.502	−0.765**	−0.574	−0.510	−0.671*	−0.941***	−0.858***	−0.766***	−0.710***
	(0.443)	(0.295)	(0.394)	(0.405)	(0.354)	(0.290)	(0.267)	(0.286)	(0.209)
Low Risk	0.094	−0.136	−0.124	−0.027	−0.079	−0.105	0.110	−0.036	0.175**
	(0.109)	(0.110)	(0.118)	(0.117)	(0.083)	(0.102)	(0.080)	(0.072)	(0.078)
Momentum	−0.121	−0.007	0.020	−0.044	−0.006	−0.017	−0.206***	−0.086	−0.204***
	(0.127)	(0.086)	(0.096)	(0.109)	(0.080)	(0.095)	(0.075)	(0.070)	(0.073)
Profit Growth	−0.281	−0.647***	−0.823***	−0.599**	−0.312	−0.359*	0.170	−0.245	−0.204
	(0.238)	(0.203)	(0.234)	(0.241)	(0.212)	(0.195)	(0.156)	(0.155)	(0.162)
Profitability	−0.073	−0.031	0.067	0.164	0.294	0.647***	−0.248	0.194	0.162
	(0.262)	(0.208)	(0.200)	(0.241)	(0.192)	(0.199)	(0.215)	(0.194)	(0.137)
Quality	0.863**	0.636***	0.521**	0.536**	0.390	0.202	0.508***	0.454	0.516***
	(0.346)	(0.162)	(0.214)	(0.240)	(0.249)	(0.271)	(0.172)	(0.301)	(0.187)
Seasonality	−0.506	−0.744***	−0.655**	−0.619**	−0.390	−0.423	−0.080	−0.435*	−0.435*
	(0.360)	(0.248)	(0.305)	(0.299)	(0.283)	(0.283)	(0.212)	(0.251)	(0.261)
Short-Term Reversal	0.202	0.195*	0.128	−0.054	−0.036	−0.035	0.002	0.149	0.044
	(0.165)	(0.116)	(0.135)	(0.138)	(0.125)	(0.140)	(0.100)	(0.101)	(0.112)
Size	0.144	0.194	0.196	0.183	0.280**	0.454***	−0.053	0.260***	0.275***
	(0.124)	(0.130)	(0.153)	(0.146)	(0.136)	(0.125)	(0.111)	(0.099)	(0.083)
Value	−0.429	−0.421*	−0.233	−0.360	−0.497*	−0.788***	−0.581**	−0.527***	−0.762***
	(0.359)	(0.253)	(0.303)	(0.311)	(0.270)	(0.270)	(0.229)	(0.196)	(0.195)
Constant	0.017	0.018	0.019	0.022*	0.012	0.006	0.019*	0.025**	0.004
	(0.014)	(0.014)	(0.014)	(0.013)	(0.012)	(0.017)	(0.011)	(0.011)	(0.009)
Average Ret.	0.037	0.019	0.025	0.033	0.017	0.008	0.028	0.02	0.011
Adj R-squared	0.352	0.281	0.315	0.326	0.192	0.169	0.320	0.218	0.311

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Same as Table 22, but for value-weighted portfolios.

nancing (red), Shareholder Payout & Capital Return Policy (gray), Pharmaceutical Development (pink), Corporate Communication (orange), Sell-Side Recommendations & Target Price Revisions (light gold), and Securities Litigation & Shareholder Investigations (light blue-gray). Internet Appendix D.6 reports ten randomly selected news summaries from each topic to illustrate their thematic content. Figure 11 visualizes the topic structure in a two-dimensional space using t-SNE. Each point represents a document (or text segment), and colors denote the assigned topic labels. The embedding maps high-dimensional semantic representations into two dimensions such that documents with similar content are positioned closer together. The clusters are spatially compact and largely non-overlapping, indicating

Figure 11: Two-Dimensional t-SNE Visualization of News Topic Clusters



Notes: This figure displays the two-dimensional t-SNE projection of news summary embeddings. Each point represents a news article and is colored according to its assigned hierarchical topic. Clustering is performed only on news associated with stocks that enter the daily long–short trading portfolio.

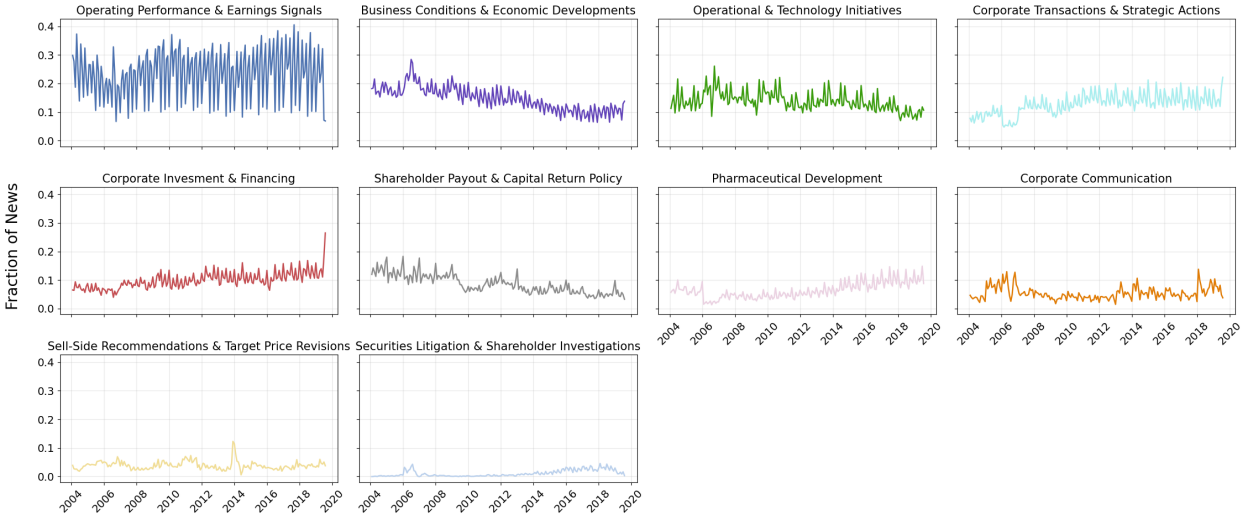
strong topical separation in the underlying embedding space.

Several patterns are evident. Operating Performance & Earnings Signals (blue) occupies a relatively large and dense region, reflecting either a higher document frequency or broader semantic variation within that category. Corporate Investment & Financing (red) and Corporate Transactions & Strategic Actions (cyan) are adjacent but remain clearly segmented, consistent with their conceptual proximity in corporate finance. Securities Litigation & Shareholder Investigations (light blue-gray) and Corporate Communication (orange) appear in smaller, more localized regions toward the periphery, suggesting more specialized language patterns. Shareholder Payout & Capital Return Policy (gray) forms a compact and distinct cluster, separated from operational themes, in line with its more specific financial-policy vocabulary.

Next, we examine the time-series evolution of news composition across topics. For each month, we compute the cross-sectional share of articles assigned to each topic among the subset of news that our model predicts to trade. Because this sample consists only of news items selected based on their predicted returns, these shares should not be interpreted as measures of aggregate media attention. Rather, they capture the relative prevalence of topics that contribute most strongly to cross-sectional return predictability at a given point in time.

The resulting patterns in Figure 12 align closely with observable corporate and financial market developments. Earnings-related operating performance news constitutes the

Figure 12: Monthly Composition of Traded News by Topic



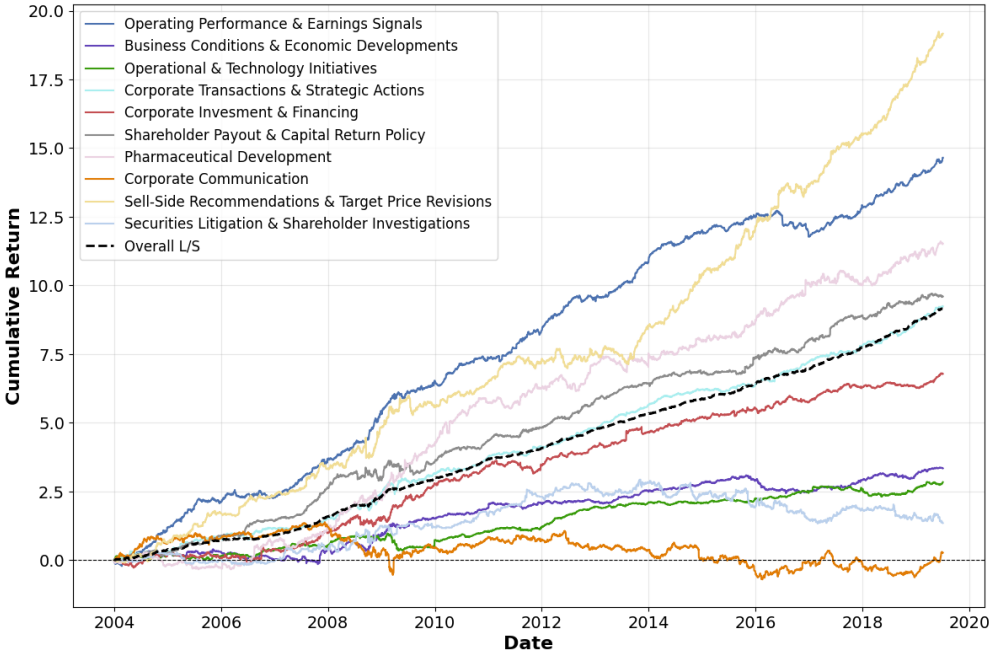
Notes: The figure shows the monthly fraction of news articles associated with each semantic topic among the set of traded stocks. For each month, we compute the cross-sectional share of articles assigned to each topic based on the LLM embedding and clustering procedure.

largest share and exhibits clear seasonality aligned with quarterly reporting cycles. This prominence is economically intuitive. Earnings announcements introduce new, structured information into the market rather than merely reflecting prior price movements. They report explicit earnings surprises—realizations relative to analyst forecasts—that directly revise expectations about future cash flows and cannot be inferred from past prices alone. In addition, earnings releases contain forward-looking guidance and managerial commentary that expand the information set available to investors. Consistent with this mechanism, the well-documented Post-Earnings Announcement Drift (PEAD) shows that returns continue to adjust after disclosure, indicating gradual incorporation of these low-frequency, fundamental information shocks.

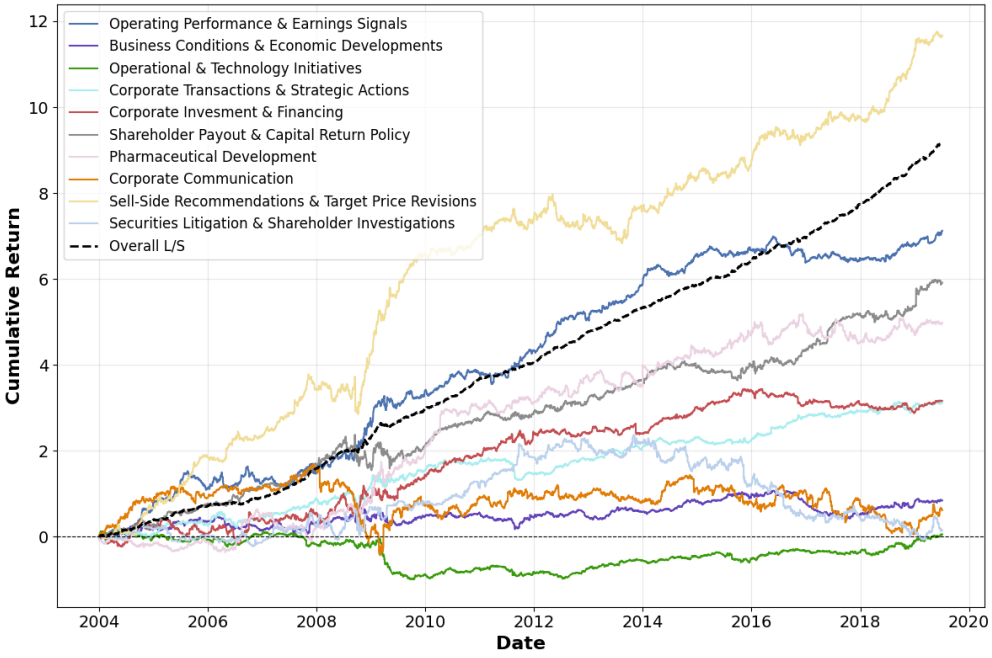
Earnings-related information features prominently in the topic decomposition. In Section 3.3, we show that excluding news within a narrow window around earnings announcement dates leaves performance largely intact, yet the earnings cluster emerges as the largest topic in the present analysis.

In the data, 44% of articles assigned to the earnings cluster occur within three days of the formal announcement date, implying that the majority fall outside this narrow window. These include previews, guidance updates, analyst commentary, and post-announcement discussions. Removing the entire earnings cluster reduces the equal-weighted Sharpe ratio (based on news summary) from 5.42 to 4.55, indicating that earnings news contributes meaningfully but does not solely account for the overall predictability.

Figure 13: Topic-Level Cumulative Long–Short Portfolio Performance



(a) Equal-Weighted Portfolios



(b) Value-Weighted Portfolios

Notes: The figure plots cumulative returns of topic-partitioned long–short portfolios constructed from stocks in the top and bottom quintiles of model-predicted returns. Each topic portfolio represents a decomposition of the baseline strategy based on the associated news cluster. Panel A reports equal-weighted returns and Panel B reports value-weighted returns. The dashed line denotes the baseline long–short portfolio.

Other topic dynamics are similarly consistent with economic intuition. Pharmaceutical development news displays a persistent upward trend over time, consistent with the growing economic importance of biotechnology and innovation-driven sectors. News classified under Securities Litigation & Shareholder Investigations shows pronounced spikes during periods of market stress—most notably around 2007–2008—reflecting heightened legal scrutiny and shareholder disputes during the global financial crisis.

We next decompose the baseline long–short strategy into topic-specific components. On each trading day, we assign stocks in the long and short legs to topics based on their associated news clusters and compute cumulative returns separately. Since clustering is conducted only on stocks already entering the baseline portfolio, these topic portfolios represent partitions of the same underlying signal rather than independent trading strategies. The results are reported in Figure 13, where the upper panel presents equal-weighted performance and the lower panel reports value-weighted results.

Substantial cross-topic heterogeneity is evident in both the portfolio return figures and the corresponding performance table. The strongest and most stable performance is delivered by the Sell-Side Recommendations & Target Revisions category. In the reported equal-weighted and value-weighted specifications, this topic consistently dominates the baseline strategy. The evidence indicates that analyst rating changes and target price revisions constitute the most robust information channel in the model. Economically, this is unsurprising: analyst reports provide explicit, forward-looking valuation updates that map directly into expected returns. Their structured format facilitates both model extraction and gradual market adjustment.

Other categories also generate economically meaningful predictability, but with weaker robustness. Operating Performance & Earnings Signals, Pharmaceutical Development, and Shareholder Payout & Capital Return Policy outperform the baseline in equal-weighted portfolios. However, their relative advantage attenuates under value weighting. The equal-weighted dominance, together with the value-weighted attenuation, suggests that part of the predictive power is concentrated in smaller firms, where information diffusion may be slower and underreaction more pronounced.

A unifying feature of the higher-performing categories is the presence of economically discrete expectation updates that translate directly into valuation changes. Analyst revisions provide explicit forecasts, earnings announcements update fundamentals in a quantifiable manner, and payout policy changes alter expected shareholder distributions. These information types are tightly linked to cash-flow expectations and therefore produce strong cross-sectional return forecasts.

In contrast, broader qualitative themes—such as corporate strategy narratives, manage-

ment commentary, investor meetings, and general operational or risk discussions—tend to underperform or track the baseline in both weighting schemes. Their cumulative return paths are flatter and less persistent. These articles typically convey diffuse or interpretative information that is harder to map into immediate valuation adjustments, resulting in weaker short-horizon signals.

In addition, part of this traded news may reflect noise in sentiment measurement that ideally would have been filtered out. For instance, routine corporate communications do not appear to convey strong directional sentiment, as illustrated by the sampled articles in the Internet Appendix. These articles are also relatively less prevalent, as shown in Figure 12, and contribute little incremental return predictability—even under equal weighting—as evidenced by Figure 13.

Overall, the decomposition demonstrates that LLM-based predictability is not uniform across news content. It is strongest when news delivers well-defined expectation updates with direct valuation consequences, and weaker when the information is qualitative or economically diffuse.

5 Conclusion

In this study, we use large language models to extract contextualized representations from financial news and examine their implications for return predictability across 16 equity markets and 13 languages. We document that financial news contains economically meaningful information that is incorporated into prices gradually rather than instantaneously. News-based signals predict short-horizon return continuation, contrasting with the reversal patterns commonly observed in raw daily returns.

Across markets and specifications, LLM-based predictors consistently outperform traditional word-based approaches, particularly when predictive content depends on richer semantic structure and contextual coherence. Performance improves with textual richness for expressive models, and AI-generated summaries often deliver the strongest and most stable results. These findings suggest that distilling salient information into a structured form enhances signal extraction, especially when paired with high-capacity language representations.

Our topic decomposition further shows that return predictability is concentrated in economically interpretable themes, including earnings-related disclosures and analyst revisions, while broader corporate narratives and routine communications contribute little incremental signal. Earnings-related content constitutes the largest thematic cluster, yet the predictive power is not confined to narrow announcement windows, nor is it driven solely by earnings

news.

We also address concerns about look-ahead bias by extending the sample beyond model knowledge cutoff dates and by implementing a structured anonymization procedure to mitigate potential information leakage. The persistence of strong predictive performance under these safeguards supports the validity of the empirical findings.

Overall, our results highlight the value of LLM-based representations for extracting economically relevant information from unstructured text in global equity markets. Beyond predictive gains, the framework provides a structured way to analyze how different types of news content are processed and incorporated into prices across markets, languages, and firm characteristics. As language models continue to evolve, integrating textual representations with models of trading behavior, market microstructure, and strategic information production offers a promising avenue for advancing empirical asset pricing and market efficiency research.

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Supplementary materials of "Expected Returns and Large Language Models"

Yifei Chen*
Booth School of Business
University of Chicago

Bryan Kelly†
Yale University, AQR Capital Management, and NBER

Dacheng Xiu‡
Booth School of Business
University of Chicago and NBER

Abstract

This Internet Appendix presents supplementary methodological details, along with additional tables and figures that support and extend the analysis in the main text.

*Address: 5807 S Woodlawn Avenue, Chicago, IL 60637 USA. Email: yifei.chen@chicagobooth.edu.
†Address: 165 Whitney Ave, New Haven, CT 06511 Email: bryan.kelly@yale.edu.
‡Address: 5807 S Woodlawn Avenue, Chicago, IL 60637, USA. Email: dacheng.xiu@chicagobooth.edu.

A Specification of Large Language Models

Table IA1 summarizes the evolution of the LLaMA model family across successive releases, highlighting differences in scale and functionality that are relevant for text representation. The table documents the available parameter sizes, the presence of instruction-tuned (chat) variants, and the introduction of multimodal capabilities in later versions. In our empirical analysis, we focus on LLaMA1-13B, LLaMA2-13B, and LLaMA3-8B, which strike a balance between model capacity and computational feasibility and facilitate meaningful comparisons across generations while keeping model scale broadly comparable. For each LLaMA generation, we also evaluate alternative model sizes and find their empirical performance to be qualitatively similar; these results are therefore omitted, indicating that our main conclusions are not sensitive to the specific model choice.

Table IA1: Comparison of LLaMA Model Versions

	LLaMA1	LLaMA2	LLaMA3	LLaMA3.1	LLaMA3.2	LLaMA3.3
# of Parameters	7B, 13B, 33B, 65B	7B, 13B, 70B	8B, 70B	8B, 70B, 405B	1B, 3B, 11B, 90B	70B
Base	✓	✓	✓	✓	✓	✓
Chat/Instruct	✗	✓	✓	✓	✓	✓
Multimodal	✗	✗	✗	✗	✓	✗

Note: This table summarizes successive versions of the LLaMA model family. Checkmarks (✓) indicate the availability of a given variant, while crosses (✗) denote its absence. The table reports model parameter sizes and key features, including base models, instruction-tuned (chat) variants, and multimodal capabilities.

Table IA2 outlines the language-specific NLP infrastructure used for BERT- and RoBERTa-based text representations. To ensure high-quality text representation across languages, we do not rely on a one-size-fits-all model. For word-based benchmarks, we employ language-specific spaCy tokenizers tailored to each market. For non-English contextual embeddings, we employ specialized monolingual BERT models when available (e.g., *bert-base-chinese* for Chinese and *bert-base-french-europeana-cased* for French), along with the multilingual xlm-roberta-large architecture to provide consistent coverage across foreign-language markets.

B SESTM Training

SESTM differs from other prediction methodologies we study in that it introduces distinctive structural assumptions on the BoW representation of article i at time t, denoted as $d_{i,t}$, and

Table IA2: Specification of Tokenizers

	BOW/Word Embeddings	BERT	RoBERTa
US	en_core_web_sm	bert-large-uncased	roberta-large
UK	en_core_web_sm	bert-large-uncased	roberta-large
Australia	en_core_web_sm	bert-large-uncased	roberta-large
Canada	en_core_web_sm	bert-large-uncased	roberta-large
China (HK)	jieba	ckiplab/bert-base-chinese	xlm-roberta-large
Japan	nagisa	cl-tohoku/bert-base-japanese	xlm-roberta-large
Germany	de_core_news_sm	bert-base-german-cased	xlm-roberta-large
Italy	it_core_news_sm	dbmdz/bert-base-italian-cased	xlm-roberta-large
France	fr_core_news_sm	dbmdz/bert-base-french-europeana-cased	xlm-roberta-large
Sweden	sv_core_news_sm	KB/bert-base-swedish-cased	xlm-roberta-large
Denmark	da_core_news_sm	Maltehb/danish-bert-botxo	xlm-roberta-large
Spain	es_core_news_sm	dccuchile/bert-base-spanish-wwm-uncased	xlm-roberta-large
Finland	sv_core_news_sm	TurkuNLP/bert-base-finnish-cased-v1	xlm-roberta-large
Portugal	pt_core_news_sm	neuralmind/bert-base-portuguese-cased	xlm-roberta-large
Greece	el_core_news_sm	nlpaueb/bert-base-greek-uncased-v1	xlm-roberta-large
Netherlands	nl_core_news_sm	wietsedv/bert-base-dutch-cased	xlm-roberta-large

Note: This table reports the specifications of pre-trained models used for each country. The column "BOW/-Word Embeddings" lists the spaCy tokenizers employed, while the "BERT" and "RoBERTa" columns report the corresponding pre-trained models obtained from Hugging Face.

its associated sentiment score $p_{i,t}$:

$$\mathrm{P}(sgn(y_{i,t}) = 1) = g(p_{i,t}), \quad d_{[S],i,t} \sim Multinomial(s_{i,t}, p_{i,t}O_+ + (1 - p_{i,t})O_-),$$

where $g(\cdot)$ is some increasing function, $s_{i,t}$ is the total number of sentiment charged words for article i at time t, O_+ and O_- are $|S| \times 1$ vectors of parameters. The set S is subset of the vocabulary with an exclusive list of sentiment-charged words, and $d_{[S],i,t}$ is a subvector of $d_{i,t}$ with rows corresponding to words in set S. [2] proposes this model and suggests that SESTM's training involves the construction of in-sample estimates for various variables. To be more specific, we construct $\widehat{S}$ by screening words based on their frequency in positive articles and construct $\widehat{O}_\pm$ by running regressions of sentiment word frequencies of each article onto pilot estimates of in-sample sentiment scores. Based on the estimated $\widehat{O}_\pm$ and $\widehat{S}$, we estimate the sentiment score for an out-of-sample article using maximum likelihood estimation. Several tuning parameters will be involved in the process, including three in the screening step, and one shrinkage parameter in the (penalized) likelihood estimation step.

C Text Anonymization Procedure

We implement a structured anonymization pipeline to mitigate potential information leakage from the raw text. The procedure replaces explicit company names, masks product and brand identifiers, standardizes references to individuals, anonymizes calendar-year and

other temporal information, replaces abbreviations and shorthand expressions, and resolves references to firms with similar or confusable names. This process ensures that the resulting text preserves semantic content relevant for prediction while removing identifiable entities and time-specific cues prior to model input.

Table IA3: Example Transformations

Function	Before	After
Products	Microsoft introduces new **Windows** feature.	Microsoft introduces new **SOME_PRODUCT** feature.
Same Company Names	**Microsoft** corp (**MSFT.O**) announced earnings.	**SOME_COMPANY** corp (**SOME_COMPANY**) announced earnings.
Similar Company Names	**Microsft** is rumored to launch a new device.	**SOME_COMPANY** is rumored to launch a new device.
Person Names	**Satya Nadella**, CEO of Microsoft, introduced ...	**SOME_NAME**, CEO of Microsoft, introduced ...
Year Info	The company expects growth in **2022**.	The company expects growth in **SOME_YEAR**.
Abbreviations	**G.S.** has announced a new investment initiative.	**SOME_COMPANY** has announced a new investment initiative.

Note: This table provides examples of the transformations applied to the text data during the anonymization process, including replacing company products, names, year information, and abbreviations.

We update the methodology of [1] by replacing the originally used GPT-3—now deprecated—with contemporary large language models, specifically GPT-4o and LLaMA-3.1-8B-Instruct.

We further refine the evaluation prompt. The baseline prompt used in [1] (Prompt 1) instructs the model to return "Unknown" when uncertain, which we find can inflate anonymization success rates by discouraging inference.[1] To conduct a more stringent stress test, we introduce an alternative prompt (Prompt 2) that explicitly encourages the model to leverage contextual and domain knowledge, mimicking the behavior of a skilled analyst:

> *"You are a financial analyst with expertise in recognizing companies based on news context, even when their names are masked. If you have potential leads regarding the company in question, output the company name without additional text. If you cannot identify the company, simply output 'unknown'."*

As reported in Table IA4, Prompt 1 consistently yields higher masking success rates than Prompt 2 across both headlines and body text. For example, when applied to headlines, the masking success rate under GPT-4o declines from 97.68% with Prompt 1 to 95.85% with Prompt 2, with a similar reduction under LLaMA. By construction, the masking success rate measures the fraction of texts for which the evaluator fails to recover masked identifying information. Consequently, lower success rates indicate a larger share of texts that remain vulnerable to re-identification and therefore exhibit greater potential information leakage.

[1]Prompt 1: "You will receive a business description. If you know to which company the description belongs, return the company name; otherwise, return 'Unknown'."

Under this interpretation, Prompt 2 provides a more conservative evaluation of anonymization quality by flagging more instances of residual leakage than Prompt 1. Moreover, for a given prompt, LLaMA3.1 8B Instruct consistently reports lower masking success rates than GPT-4o, reflecting its stronger ability to recover masked entities. We therefore adopt LLaMA3.1 8B Instruct with Prompt 2 as the benchmark evaluation procedure, which minimizes the risk of understating information leakage and the associated look-ahead bias.

Table IA4: Comparison of Masking Success Rates by Evaluation Prompt

	Headline		Body	
	LLaMA	GPT	LLaMA	GPT
Prompt 1	97.58%	97.68%	86.40%	90.80%
Prompt 2	95.13%	95.85%	76.43%	79.00%

Note: This table reports the masking success rates of the pattern-based method evaluated on headlines and body text using two different prompts (Prompt 1 vs. Prompt 2). Evaluations are conducted using LLaMA3.1 8B Instruct and GPT-4o.

Table IA5: Performance Statistics of Pattern-Based Masking

	Articles	Alerts	Both
# Sample	3,000	3,000	6,000
Mask Success Rate (%)	76.43	95.20	85.82
Unknown Rate (%)	13.47	26.20	19.83

Note: This table reports the effectiveness and efficiency of the pattern-based masking method. Results are based on a random sample of 3,000 news alerts and 3,000 news articles from 1996 to 2019, evaluated with LLaMA3.1 8B Instruct using Prompt 2.

D Additional Empirical Details and Results

D.1 International Equity Markets: Summary Statistics

Table IA6 provides summary statistics for the 16 equity markets in our sample, along with a consolidated non-U.S. dataset. The table reports the primary language of news coverage, local market trading hours, the total number of processed news articles, sample start dates, the average number of stocks, the number of trading days, and the average daily volume of news. Sample start dates vary by data availability, ranging from the mid-1990s for major economies such as the United States, United Kingdom, Japan, and Germany, to the early-to-mid 2000s for smaller markets. The "non-US" row aggregates statistics across all international markets excluding the United States.

D.2 International News Coverage and Filtering

Table IA7 summarizes the filtration pipeline applied to the raw international news corpus. Starting from all available articles obtained from Thomson Reuters Real-Time News Feed (RTRS) and third-party archives (3PTY), we apply a sequence of filters designed to isolate firm-level, idiosyncratic news. These filters restrict attention to articles that can be unambiguously mapped to a single stock, matched to valid return data, and that satisfy length and similarity criteria. The table reports the number of remaining articles after each stage of the filtering process, illustrating the extent of the reduction from the raw corpus to the final set of news used in the empirical analysis.

Table IA6: Summary Statistics of International Markets

	Language	Market Hours	# of Articles	Initial Day	Avg. # of Stocks	# of Days	Avg. # of News
US (Alert)	English	09:30 – 16:00	2,935,852	1996-01-02	1,746	5,929	10,337
US	English	09:30 – 16:00	3,038,025	1996-01-02	2,593	5,933	10,697
UK	English	08:00 – 16:30	571,285	1996-01-02	454	6,087	2,011
Australia	English	10:00 – 16:00	249,190	1996-01-03	287	6,033	877
Canada	English	09:30 – 16:00	350,549	1996-01-03	406	6,032	1,234
China (HK)	Chinese	09:30 – 16:00	182,363	1996-01-03	247	5,768	642
Japan	Japanese	09:00 – 15:00	310,244	1996-01-05	645	5,875	1,092
Germany	German	09:00 – 17:30	178,039	1996-01-03	163	6,031	626
Italy	Italian	09:00 – 17:30	130,168	1996-01-05	97	5,778	458
France	French	09:00 – 17:30	153,779	1996-01-03	167	5,994	541
Sweden	Swedish	09:00 – 17:25	115,195	2001-06-07	170	4,629	526
Denmark	Danish	09:00 – 16:55	43,584	1996-01-22	37	4,559	156
Spain	Spanish	09:00 – 17:30	34,159	1996-01-05	37	5,520	120
Finland	Finnish	10:00 – 18:25	28,633	2003-01-03	50	4,025	143
Portugal	Portuguese	11:30 – 16:30	6,158	2005-05-13	11	2,616	36
Greece	Greek	10:15 – 05:20	7,710	2003-02-19	16	3,057	39
Netherlands	Dutch	09:00 – 17:30	3,751	2005-09-20	11	2,102	22
non-US			2,364,807		2,735	6,147	8,326

Note: This table summarizes market information and processed datasets for each country. The "non-US" row reports summary statistics for all international markets excluding the US

D.3 Trade Execution Timing and Portfolio Performance

This subsection examines how trade execution timing affects portfolio performance. Our models are trained using opening prices and generate predictions for open-to-open (O2O) returns, which conceptually correspond to trading at the market open. In practice, however, executing trades precisely at the open may be difficult due to liquidity, operational, or market impact constraints. To assess the robustness of our results to alternative execution schemes, we retain the same O2O return predictions but rebalance portfolios and evaluate performance using alternative return definitions based on volume-weighted average prices (VWAP) and close-to-close (C2C) returns. VWAP captures execution distributed throughout the trading day, while C2C represents the latest feasible execution within the same trading session.

Table IA7: International News Summary Statistics

	Raw Articles			Articles Tagged with Single Stock			Articles With Returns	After Filtering Short Articles	After Filtering Similarity > 0.8
	RTRS	3PTY	Total	RTRS	3PTY	Total			
US	6,366,019	4,843,867	11,209,886	2,863,166	4,123,823	6,986,989	4,755,247	4,123,279	3,038,025
UK	707,288	1,050,467	1,757,755	196,573	773,266	969,839	906,705	901,838	571,285
Australia	261,020	1,203,784	1,464,804	100,444	1,113,347	1,213,791	388,585	382,114	249,190
Canada	255,933	473,686	729,619	126,281	431,401	557,682	481,891	478,205	350,549
China (HK)	3,537,487	7,287,688	10,825,175	1,140,542	5,558,763	6,699,305	2,086,045	305,335	182,363
Japan	3,259,103	38,860	3,297,963	1,210,077	16,850	1,226,927	405,341	399,185	310,244
Germany	2,423,671	1,751,231	4,174,902	480,264	880,650	1,360,914	238,577	229,265	178,039
Italy	1,022,204	337,322	1,359,526	194,650	227,599	422,249	173,250	168,410	130,168
France	2,422,338	1,587,490	4,009,828	298,886	670,469	969,355	174,917	174,784	153,779
Sweden	288,395	189,424	477,819	96,039	124,862	220,901	126,211	126,168	115,195
Denmark	261,146	124,209	385,355	93,596	57,768	151,364	53,056	52,381	43,584
Spain	2,748,601	165,468	2,914,069	257,739	46,829	304,568	47,541	45,597	34,159
Finland	81	110,123	110,204	38	87,226	87,264	38,159	38,119	28,633
Portugal	747,069	39,086	786,155	124,017	13,638	137,655	11,265	11,212	6,158
Greece	85,915	14	85,929	19,156	6	19,162	10,093	10,082	7,710
Netherlands	194	183,668	183,862	53	66,669	66,722	4,313	4,312	3,751

Note: This table reports the sample size after applying successive filters to news articles in international equity markets. The "Raw Articles" column shows the number of available articles from RTRS and 3PTY. The "Articles Tagged with Single Stock" columns indicate the number of articles linked to a single stock. The "Articles with Available Returns" column reports the number of articles remaining after matching with returns data. The "After Filtering Short Articles" column includes articles with lengths between 100 and 100,000 characters.

Table IA9 reports portfolio performance across these execution strategies. In the equal-weighted specification (Panel A), the O2O portfolio exhibits the strongest performance, delivering a long–short return of 0.50 with a Sharpe ratio of 4.59. Performance deteriorates as execution is delayed: trading at VWAP or at the close reduces the Sharpe ratio to 2.07 and 2.19, respectively. This decay is even more pronounced for value-weighted portfolios (Panel B), which place greater weight on large-cap stocks. While the O2O strategy remains profitable (Sharpe ratio of 1.10), performance under close-to-close execution nearly vanishes (Sharpe ratio of 0.14). These results indicate that news is incorporated into prices rapidly for large, liquid stocks, with most of the predictive content dissipating within a single trading day, whereas smaller stocks exhibit slower price adjustment and more persistent return predictability, consistent with the findings in the main text.

D.4 Model Scale and Portfolio Performance

Table IA8 reports Sharpe ratios for portfolios constructed using different LLaMA model variants. Overall, performance is broadly similar across model generations and parameter sizes, with no monotonic relationship between model scale and Sharpe ratios. Although larger and more recent models—most notably LLaMA2-70B—often deliver higher Sharpe ratios, these gains are uneven across weighting schemes, text representations, and alert subsamples.

Alert-based portfolios generally outperform article-based portfolios in the equal-weighted specification, particularly when focusing on early alert sequences (TS1 and TS2). Taken together, these findings suggest that while model scale and alert timing can matter, the

incremental improvements are modest and not fully robust. A plausible interpretation is that once the language model is sufficiently sophisticated to extract the relevant semantic content, further gains are constrained by the intrinsically low signal-to-noise ratio in financial markets, rather than by limitations in processing or understanding news text.

Table IA8: Sharpe Ratios Comparison between Portfolios Built on Variants of LLaMA Model

	EW					VW				
	Article	Alert				Article	Alert			
		All	TS1	TS2	Rest		All	TS1	TS2	Rest
LLaMA7B	3.99	3.51	3.82	2.69	2.03	0.75	1.17	1.37	1.25	1.10
LLaMA13B	4.04	4.23	3.58	2.94	2.13	0.89	1.56	1.25	1.03	0.57
LLaMA2 7B	4.25	4.18	4.33	3.11	2.58	0.84	1.60	1.74	0.99	1.31
LLaMA2 13B	4.36	4.67	3.59	2.72	2.55	1.02	1.63	1.07	0.92	0.56
LLaMA2 70B	4.89	5.04	5.06	4.02	3.53	1.09	1.85	2.23	1.03	1.44
LLaMA 8B	4.59	4.79	3.79	3.02	2.72	1.10	1.68	1.54	0.83	0.84
LLaMA 70B	3.98	3.95	4.00	3.11	2.55	1.03	1.56	1.43	1.41	1.18

Note: This table reports Sharpe ratios of quintile long–short portfolios constructed using different LLaMA model variants. The models include LLaMA (7B and 13B parameters), LLaMA2 (7B, 13B, and 70B), and LLaMA3 (8B, baseline, and 70B). Columns labeled "TS1" and "TS2" use only alerts from take sequence 1 and 2, respectively; "Rest" uses the remaining alerts; and "All" uses all available alerts.

Table IA9: Trade Execution Timing and Portfolio Performance

Panel A: Equal-Weighted Portfolios

Ret Type	O2O Portfolio			C2C Portfolio			VWAP Portfolio		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.43	-0.07	0.50	0.21	0.02	0.19	0.19	-0.04	0.23
Std	0.20	0.22	0.11	0.20	0.21	0.09	0.18	0.20	0.11
SR	2.08	-0.33	4.59	1.02	0.07	2.19	1.09	-0.21	2.07

Panel B: Value-Weighted Portfolios

Ret Type	O2O Portfolio			C2C Portfolio			VWAP Portfolio		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.22	0.09	0.12	0.09	0.08	0.02	0.12	0.07	0.05
Std	0.19	0.20	0.11	0.20	0.20	0.11	0.16	0.16	0.10
SR	1.11	0.47	1.10	0.48	0.40	0.14	0.72	0.43	0.50

Note: This table reports the performance of LLaMA3-based long–short portfolios under alternative trade execution timings. O2O denotes open-to-open returns, C2C denotes close-to-close returns, and VWAP denotes returns based on volume-weighted average prices. Panel A reports equal-weighted portfolios, while Panel B reports value-weighted portfolios.

D.5 Impact of Context Length Truncation

To ensure a consistent comparison across models with different maximum context lengths, we construct our baseline embeddings using the first 512 tokens of each news article for all

LLMs. We then exploit the extended context capacity of LLaMA3, which can process inputs of up to 8k tokens, to assess sensitivity to truncation. Specifically, we compare portfolio performance using progressively longer input limits—1k, 4k, and the full 8k-token context supported by LLaMA3.

The results, reported in Table IA10, show that performance under the truncated specification is comparable to, and in some cases slightly better than, that obtained using the full context. For example, the equal-weighted (value-weighted) long–short Sharpe ratio changes only modestly from 4.59 (1.10) under the baseline specification to 4.39 (0.85) when longer contexts are used. Expanding the context window from 1k to 8k tokens yields virtually identical results, indicating that performance converges quickly as the window expands and that incremental context beyond the initial portion of the article contributes little additional predictive content.

This rapid convergence is consistent with the distribution of article length in our sample. The vast majority of articles are relatively short: 60.5% contain fewer than 512 tokens and 81.9% contain fewer than 1k tokens, and only 2.5% exceed 4k characters, which implies that 512–1k token window already captures most of the economically meaningful content for most articles. As a result, extending the context window beyond 1k tokens affects only a small fraction of observations, helping explain why predictive performance stabilizes quickly.

Overall, these findings indicate that our results are robust to reasonable truncation choices.

Table IA10: Text Truncation and Portfolio Performance

	LLaMA3						LLaMA3 1k tokens					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.43	-0.07	0.50	0.22	0.09	0.12	0.41	-0.06	0.47	0.21	0.12	0.10
Std	0.20	0.22	0.11	0.19	0.20	0.11	0.20	0.22	0.11	0.19	0.20	0.11
SR	2.08	-0.33	4.59	1.11	0.47	1.10	2.00	-0.29	4.38	1.09	0.59	0.85
	LLaMA3 4k tokens						LLaMA3 8k tokens					
	EW			VW			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Ret	0.41	-0.06	0.47	0.21	0.12	0.09	0.41	-0.06	0.47	0.21	0.12	0.09
Std	0.20	0.22	0.11	0.19	0.20	0.11	0.20	0.22	0.11	0.19	0.20	0.11
SR	2.00	-0.28	4.39	1.09	0.59	0.83	2.00	-0.28	4.39	1.09	0.59	0.83

Note: This table reports portfolio performance using LLaMA3 embeddings constructed under different maximum context lengths (baseline, 1k, 4k, and 8k tokens). The baseline specification uses the first 512 tokens of each article. All returns and Sharpe ratios are annualized.

D.6 Representative News Summaries by Topic

This section provides qualitative illustrations of the topic structure identified by our clustering procedure. For each of the ten topics, we report ten representative news summaries randomly drawn from the corresponding model-identified clusters. These examples reflect the semantic content captured by the algorithm and illustrate the core informational themes within each topic.

Table IA11: Representative Text by Topic

Topic label	Representative Text
Operating Performance & Earnings Signals	1. Royal Bank of Canada expects to take a $130 million aftertax charge in the fourth quarter due to expected claims related to Hurricanes Katrina and Rita. (2005-09-29) 2. FMC Corp reported a 92% increase in quarterly revenue to $1.3 billion, beating analyst expectations, and raised its full-year earnings guidance to $5.90 to $6.20 per share. (2018-08-01) 3. Abcam Plc's trading performance has been in line with expectations, with strong sales growth in North America and more subdued growth in the UK due to uncertainty caused by the UK Government spending review. (2010-12-20) 4. Zogenix Inc reported a Q1 loss per share of $0.16, beating the Thomson Reuters IBES view of $0.14, and revenue of $18.3 million, exceeding the IBES view of $17.1 million. (2012-05-14) 5. Teledyne Technologies reported record sales of $732.5 million and GAAP earnings per diluted share of $2.32 in the second quarter, with the company raising its full year 2018 GAAP earnings outlook to $8.18 to $8.28. (2018-08-02) 6. Bigben Interactive reported a 1.8% decline in sales for the first quarter of fiscal year 2015-2016, with EUR 34.7 million in revenue, but confirmed its prospects for annual sales growth. (2015-07-27) 7. Grupa HRC SA reported a net loss of 160,606 zlotys in Q1, compared to a loss of 23,569 zlotys in the same period last year, with revenue increasing to 236,685 zlotys from 3.6 million zlotys. (2017-05-12) 8. Lincoln National Corp expects investment losses in 2010, but anticipates they will be lower than in 2009. (2010-02-09) 9. T. Clarke plc, a UK-based electrical engineering company, has reported a 6% decline in adjusted profit before interest and nonrecurring items to £7.07m for the year ended 31 December 2006, despite a strong order book worth over £200m. (2007-03-09) 10. Investors Bancorp, Inc. reported a net interest income of $159,606 for the three months ended September 30, 2016, with a net interest rate spread of 2.80% and a net interest margin of 3.00%. (2016-10-27)
Business Conditions & Economic Developments	1. Telefonica is expected to report falling revenues and OIBDA, but a surge in net profit due to writedowns on its Italian and Irish assets. (2014-02-21) 2. Progressive Insurance Company has reduced its workforce in the IT and Personal Lines groups by 341 employees, about 7% of its IT workforce and 1% of its Personal Lines group, as part of its reorganization announced in September. (2007-11-15) 3. Russian banks lowered mortgage rates by 0.3 percentage points in April, bringing the average rate to 12.6 percent since the beginning of the year, according to central bank statistics. (2013-06-06)

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Table IA11 (continued)

Topic label	Representative Text
	4. Enbridge Inc's Line 2A pipeline in Alberta remains shut following a leak, with no estimated date for restart, and the company is working with customers to mitigate potential impacts. (2017-02-20) 5. U.S. stocks fell more than 1 percent on Friday as weak job growth in December fueled worries about a U.S. recession, with the Labor Department reporting a disappointing 18,000 new jobs created, the weakest since August 2003. (2008-01-04) 6. Orica Ltd, the world's biggest explosives maker, is seeking acquisitions to expand its business in Russia, China and Africa. (2008-07-21) 7. Entergy Corp's CEO Leo Denault stated that running the Vermont Yankee nuclear plant is no longer economical due to increased capital costs and low power prices, leading to the decision to shut down the plant. (2013-08-27) 8. Ohio attorney general Richard Cordray filed a shareholder class-action lawsuit against Bank of America Corp, alleging the bank failed to disclose billions of dollars in losses and bonuses to Merrill Lynch executives and employees. (2009-09-28) 9. U.S. LNG shipments to China are expected to increase over the long term despite ongoing trade tensions, according to senior executives of Cheniere Energy LNG.A, the biggest U.S. exporter of the superchilled fuel. (2019-04-04) 10. A power outage caused by a fire at a transformer station in western Toronto has knocked out traffic lights and trading screens, but appears to have had little impact on financial markets. (2010-07-05)
Operational & Technology Initiatives	1. Moody's employees donated over 50 bicycles to the families of U.S. service members through a "Bike Build" event held in partnership with the USO of Metropolitan New York. (2013-05-23) 2. AT&T has launched a professional services package called "AT&T Consulting Essentials" designed to help small and medium-sized businesses plan, install, and deploy IT networks, as well as implement industry best practices for information security. (2006-10-04) 3. The Special Olympics Unified Relay Across America, presented by Bank of America, will stop in Atlanta on June 6, 2015, as part of a 46-day journey across the United States to the 2015 Special Olympics World Games in Los Angeles. (2015-06-02) 4. Constellation Software Inc. has acquired 100% of Honeywell POMS Corp.'s fully diluted shares, expanding its presence in the Manufacturing Execution System software space. (2012-12-05) 5. Dmitri Krakovsky, senior vice president for all HCM products at SuccessFactors, shares how his personal experience helping his sons with learning issues influenced his approach to cloud innovation in corporate learning tools. (2015-10-13) 6. Digital Ally Inc has received an order for 125 FirstVu HD bodyworn cameras and 18 patented Vulink connectivity systems from an unnamed customer. (2016-11-16) 7. The Principal Financial Group was named the 2004 Corporate Partner of the Year by the National Association of Women Business Owners (NAWBO) for its support of women business owners. (2004-06-07) 8. SolarWinds, a leading provider of IT management software, will exhibit at CiscoLive! 2014 in Milan to connect with networking professionals and demonstrate its products, including SolarWinds NetFlow Traffic Analyzer, SolarWinds Network Performance Monitor, and SolarWinds Network Configuration Manager. (2014-01-20) 9. France Télévisions is using Avid's postproduction workflow to produce daily highlights coverage of the 2015 Tour de France, relying on a powerful and highly collaborative workflow built on the Avid MediaCentral Platform. (2015-07-07)

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Table IA11 (continued)

Topic label	Representative Text
	10. ATT has announced the panel of judges for its 2009 Big Mobile on Campus Challenge, a contest that recognizes innovative mobile applications that enhance academic performance and campus operations, with a grand prize of $10,000 and a trip to the EduCause Annual Conference. (2009-07-28)
Corporate Transactions & Strategic Actions	1. Digital Realty Trust Inc. acquired a data center facility in Dublin, Ireland for 36.3 million euros and started construction on a new 120,000 square foot data center on land it acquired in 2006. (2007-02-05) 2. Duke Energy Corp's Carolinas unit is seeking bids for a $100 million North Carolina solar plan that aims to install photovoltaic solar panels at up to 850 sites, generating at least 16 megawatts of electricity. (2008-09-04) 3. Bank of Montreal has confirmed the election of its board of directors, with all nominees elected. (2019-05-03) 4. Domino's Pizza Inc. announced the departure of its Chief Marketing Officer, Ken Calwell, and said its USA President, Patrick Doyle, will fill the position on an interim basis while the company searches for a new CMO. (2008-05-12) 5. Spectrum Brands Inc. plans to close its acquisition of Russell Hobbs this summer after finding no superior offers during its "goshop" period. (2010-04-12) 6. Comerica Incorporated has appointed Patricia McCann as vice president and national Civic Affairs manager to implement a volunteer program to increase employee involvement in all markets. (2008-02-26) 7. Axalta Coating Systems is expanding its presence in India with investments that will double production capacity and establish an automotive OEM coatings development laboratory at its operations center in Savli, Gujarat State. (2015-10-12) 8. Camtek Ltd. has appointed Nir Dery as VP marketing and Gilad Golan as VP Research and Development Division, both of whom have extensive experience in the semiconductor industry. (2010-01-05) 9. Comstock Holding Companies, Inc. has partnered with Stratford Capital Group to develop 110 upscale workforce housing units at the Shady Grove Metro Station in Washington D.C. (2016-11-14) 10. GE Capital has underwritten $500 million of working capital facilities to support the future requirements of Alcan Engineered Products, a leading global aluminium products manufacturer, in a deal that supports the creation of a new joint venture. (2010-08-09)
Corporate Invesment & Financing	1. Superconductor Technologies Inc. has priced a public offering of 2,571,429 shares of stock and warrants to purchase an additional 2,571,429 shares, with expected total gross proceeds of approximately $9 million. (2018-07-26) 2. Navistar International Corp has renewed a $500 million dealer inventory funding facility, which is funded through two major relationship banks. (2011-07-20) 3. Catalyst Media Group PLC has issued 14.7 million new shares to Champ Car World Series LLC, representing 8.28% of its share capital, in exchange for Champ Car's agreement to pay up to £1.14 million in financial obligations of its television production subsidiary. (2005-01-05) 4. Xtant Medical Holdings has completed an accounts receivable revolver credit facility with Silicon Valley Bank, providing up to $6 million in funds. (2016-05-26) 5. Barrick Gold Corporation has been granted a court order allowing its Cortez Hills Mine to continue operations while a study is conducted on three aspects of the project, preserving hundreds of Nevada mining jobs. (2010-04-14) 6. CoreMark Holding Company, Inc. has amended its $200 million credit facility, reducing the margin on LIBOR borrowings and unused facility fees, and extending the term for two years. (2013-05-30)

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Table IA11 (continued)

Topic label	Representative Text
	7. Paynova AB proposes an Extraordinary General Meeting (EGM) to issue up to 53.3 million new shares at 0.40 Swedish crowns per share, with existing shareholders having preferential rights to subscribe. (2015-09-22) 8. BlackRock, Inc. has acquired a 3.71% stake in Stabilus S.A. through the purchase of shares and the exercise of conversion rights, bringing its total voting rights to 5.73%. (2015-12-09) 9. Private Commercial Finance Group Plc held its Annual General Meeting and all resolutions were passed. (2008-10-24) 10. Bangkok Bank Public Company Limited has sold shares in GoldenRiver Investment International Corporation, reducing its shareholding from 10.28% to 10% of the company's total paid-up capital. (2004-12-13)
Shareholder Payout & Capital Return Policy	1. Moody's has placed the ratings of two classes of commercial mortgage-backed floating rate notes on review for possible downgrade due to decreased pool diversity and a tenant guarantor downgrade. (2006-10-24) 2. Standard & Poor's said that State Street Corp.'s ratings are not affected by its exposure to four paper conduits, citing the company's ability to absorb the assets onto its balance sheet. (2007-08-28) 3. Moody's Investors Service has downgraded HeidelbergCement's corporate family and debt issuance ratings to Ba3 from Ba1 due to increasing uncertainties related to the company's liquidity situation, which has started to address. (2008-12-03) 4. Thor Industries, Inc. announced the payment of a regular quarterly dividend of $0.27 per share, payable on April 20, 2015 to shareholders of record at the close of business on April 6, 2015. (2015-03-20) 5. Lithia Motors Inc. declared a quarterly dividend of $0.12 per share for the fourth quarter of 2005, with shareholders receiving the dividend on March 6, 2006. (2006-02-08) 6. Target Corporation has declared a quarterly dividend of 7 cents per share, marking the 146th consecutive dividend paid by the company since it became publicly held in 1967. (2004-01-15) 7. TriplePoint Venture Growth BDC Corp. has declared an initial dividend of $0.09 per share, reflecting a quarterly dividend rate of $0.30 per share, and will be payable on April 30, 2014 to stockholders of record on April 15, 2014. (2014-04-03) 8. Banc of California declared a quarterly cash dividend of $0.12 per share, with an annualized dividend yield of 3.99%, and announced that it will be the eighteenth cash dividend subject to its Dividend Reinvestment Plan. (2015-09-02) 9. W. R. Berkley Corporation declared a regular quarterly cash dividend of 13 cents per share and a special cash dividend of 50 cents per share to be paid on October 5, 2016 to stockholders of record at the close of business on September 20, 2016. (2016-08-02) 10. FRP Holdings Inc's board approved a $10 million increase in the company's stock repurchase authorization. (2018-12-06)
Pharmaceutical Development	1. INCY's JAK1 inhibitor, INCB39110, demonstrated efficacy in a Phase II trial for rheumatoid arthritis, with the highest dose tested showing the most effective results, and was generally well-tolerated with no evidence of myelosuppression or immunosuppression. (2013-10-28) 2. Agenus Inc. has secured a $115 million nondilutive royalty transaction with an investor group led by Oberland Capital Management, LLC, which will provide funding to advance its immunooncology programs. (2015-09-09) 3. Bioanalytical Systems Inc. has acquired the Smithers Avanza Toxicology Services business unit for approximately $1.2 million in cash and 200,000 common shares, expanding its scientific talent and supporting its growing toxicology business. (2019-05-01)

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Table IA11 (continued)

Topic label	Representative Text
	4. Boehringer Ingelheim and Lilly announce an academic collaboration with the University of Oxford to investigate the effects of Jardiance on the progression of kidney disease and cardiovascular death in adults with established chronic kidney disease with and without diabetes. (2018-04-16) 5. GlaxoSmithKline plans to apply for a reissued U.S. patent on asthma drug Advair to strengthen its exclusive rights to the medicine, citing concerns over the patent's language and potential loss of protection. (2004-07-27) 6. The pharmaceutical industry is facing new regulatory and recall challenges due to an influx of legitimate and counterfeit drugs entering the market, mergers and acquisitions, and regulatory shifts, making recall preparedness more important than ever. (2016-02-17) 7. Aldoxorubicin has demonstrated significantly superior efficacy over doxorubicin in prolonging progression-free survival and improving rates of 6-month progression-free survival and tumor response in patients with metastatic or locally advanced unresectable soft tissue sarcoma. (2015-09-21) 8. Cirius Therapeutics' lead drug candidate MSDC0602K has shown potential in treating non-alcoholic steatohepatitis (NASH) in preclinical data presented at The International Liver Congress, demonstrating reduced disease progression and fibrosis in a 3D bioprinted human liver tissue model. (2018-04-12) 9. Puma Biotechnology announced that the European Medicines Agency has validated its Marketing Authorization Application for neratinib as an extended adjuvant treatment for HER2-positive early stage breast cancer in Europe. (2016-08-22) 10. SCOLR Pharma has initiated human trials for its 12-hour controlled delivery technology (CDT) based pseudoephedrine tablets, with the goal of submitting an FDA application this summer. (2005-03-01)
Corporate Communication	1. Vishay Intertechnology, Inc. will present at the Banc of America Securities 34th Annual Investment Conference on September 21, 2004, and investors can listen to an audio webcast of the presentation. (2004-09-17) 2. Ford Motor Company will host a conference call on April 1, 2005 to discuss its March 2005 U.S. sales, and the call will be available via phone or webcast. (2005-03-23) 3. Heska Corporation will hold a webcast conference call to discuss its third quarter 2006 earnings on November 9, 2006, at 9:00 a.m. MST. (2006-11-09) 4. Benefitfocus, Inc. announced that its CEO, Shawn Jenkins, will present at the J.P. Morgan Technology, Media and Conference on May 25, 2016, and the presentation will be available via live webcast. (2016-05-09) 5. Novavax, Inc. announced that its CEO, Stanley C. Erck, will present at the 25th Annual ROTH Conference on March 19, 2013, and a live webcast of the presentation will be available on the company's website. (2013-03-13) 6. Vertex Pharmaceuticals will webcast its corporate presentations at several upcoming conferences, including the Bank of America and Merrill Lynch 2009 Health Care Conference and the Deutsche Bank 34th Annual Health Care Conference. (2009-05-06) 7. Berkshire Hills Bancorp, Inc. will release its second quarter 2016 earnings on July 25, 2016, and hold a conference call on July 26, 2016, to discuss the results and provide guidance on future performance. (2016-07-06) 8. MaxLinear, Inc. will present at two financial conferences in May and June, with live webcasts available for investors to access. (2014-05-05) 9. ADT Corporation will host an Analyst and Investor Day meeting on May 14, 2015, where executives will present and answer questions about the company's security and automation solutions. (2015-04-02)

Table continued on next page

Table IA11 (continued)

Topic label	Representative Text
	10. Worthington Industries will host a conference call with its Chief Financial Officer to discuss the company's recent news release, and the call will be available via live audio webcast. (2015-02-24)
Sell-Side Recommendations & Target Price Revisions	1. Goldman Sachs has downgraded Admiral Group PLC to a "neutral" rating from "buy" (2014-02-10) 2. Big Lots Inc's executive says the overall deal environment remains good, with some areas being more vibrant than others. (2008-08-26) 3. Shares of Chinese online lottery platform WBAI.N fell 11.6% to $16.95 after the company's Q4 adjusted profit missed estimates due to increased costs. (2015-02-11) 4. Shares of Agricultural REIT FPI.N fell as much as 7.8% in intraday trading, its largest drop since January, following the company's offering of 3.1 million shares at a 5% discount to its last close. (2016-11-30) 5. A bullish option strategist has placed a large three-way trade on eBay, targeting a 9.5% gain in the company's stock price over the next two months, with the trade's maximum profit being reached if eBay can rally to $34. (2011-11-10) 6. Shares in MSG Networks Inc fell as much as 11.7% after its quarterly report showed a 4% decrease in revenue compared to the year-ago quarter, despite beating Wall Street expectations for EPS and revenue. (2019-05-02) 7. Shares of The St. Joe Co. have more than doubled in the past year, driven by rising housing and land prices in Florida, and analysts expect strong earnings growth in the coming years. (2005-06-19) 8. Citigroup has upgraded its price target for Enagas to 20 euros from 18 euros, citing a capex upgrade and lower tax rate, while maintaining its "buy" rating. (2006-04-04) 9. UBS Investment Research downgraded Yelp's stock to "sell" from "neutral" citing slowing revenue growth and heightened margin pressures. (2016-03-09) 10. Under Armour's shares fell as much as 14.9% to $32.25, the biggest intraday percentage loss in nearly 8 years, due to slower sales growth in North America and disappointing earnings guidance. (2016-10-25)
Securities Litigation & Shareholder Investigations	1. A class action lawsuit has been filed against XBiotech Inc. on behalf of investors who purchased the companys securities between April 15, 2015 and November 23, 2015, alleging that the company made false and misleading statements about its business and clinical development of Xilonix. (2015-12-02) 2. Insmed, Inc. (INSM) is being sued for allegedly misleading shareholders about its lead product candidate, Arikayce, and its chances of approval by the European Medicines Agency. (2016-07-18) 3. Rowley Law PLLC is investigating potential claims for breach of fiduciary duty involving the directors and officers of QuadGraphics, Inc. (2017-09-28) 4. A shareholder class action lawsuit has been filed against On Deck Capital, Inc. alleging that the company's Initial Public Offering (IPO) contained untrue statements of material fact and omitted to state material facts required to be disclosed. (2015-08-19) 5. A federal securities class action lawsuit has been filed against Rockwell Medical, Inc. and its executives, alleging that they made false statements regarding the FDA approval of the company's main product offering, Triferic. (2016-04-12) 6. Hagens Berman Sobol Shapiro LLP is investigating possible securities law violations by AZZ, Inc. following the company's admission of improper accounting practices and its announcement of an investigation into historical financial statements. (2018-01-10) 7. A class action lawsuit has been filed against BrightView Holdings, Inc. for allegedly making false and misleading statements in its initial public offering registration statement, resulting in investor losses. (2019-04-16)

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Table IA11 (continued)

Topic label	Representative Text
	8. Law Offices of Howard G. Smith is investigating The Rubicon Project, Inc. for possible violations of federal securities laws, specifically regarding the company's initial public offering and recent executive departures. (2017-03-30) 9. Abraham, Fruchter Twersky, LLP is investigating possible violations of federal securities laws by Ebix, Inc. regarding accounting manipulations and misrepresentations of the company's business and financial results between May 6, 2009 and June 30, 2011. (2011-07-15) 10. Faruqi Faruqi, LLP is investigating the Board of Directors of NewMarket Corporation for potential breaches of fiduciary duties in connection with their conduct in seeking shareholders' approval for the NewMarket Corporation 2014 Incentive Compensation and Stock Plan. (2014-03-14)

References

[1] Breitung, C., Müller, S., 2025. Global business networks. Journal of Financial Economics 166, 104007. URL: https://www.sciencedirect.com/science/article/pii/S0304405X25000157, doi:https://doi.org/10.1016/j.jfineco.2025.104007.

[2] Ke, Z.T., Kelly, B.T., Xiu, D., 2019. Predicting Returns With Text Data. Working Paper 26186. National Bureau of Economic Research. URL: http://www.nber.org/papers/w26186, doi:10.3386/w26186.